

REACTJS

01

JavaScript Essentials & Advanced

Master the language of the web — from your first `alert()` to async APIs — so React makes complete sense from Day 1.

DURATION	LEVEL	MODULE	UPDATED
20 Sessions	Beginner	Module 1	June 2026

IN THIS MODULE

01	Introduction to JavaScript & Environment Setup
02	Variables & Data Types
03	Operators & Conditions
04	Loops & Iteration
05	Functions & Reusability
06	DOM Manipulation – Selecting Elements
07	DOM Events & Interactivity
08	Arrays & Objects – Data Structures
09	JSON & LocalStorage (Persistence Basics)
10	ES6 Syntax & Modern Features
11	Asynchronous JS – Callbacks, Promises & Async/Await
12	Asynchronous JS – Fetch API
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- 19 Mini-Project Preparation & Integration
- 20 Integration Testing & Polish

SESSION

01

60 MIN · FUNDAMENTALS

Introduction to JavaScript & Environment Setup

◆ Learning Objectives

- Explain what JavaScript is and why it is the language of the web
- Distinguish between client-side and server-side execution
- Describe how a browser reads and runs a JavaScript file
- Connect a JS file to an HTML page using the `<script>` tag
- Run JavaScript in the browser console and in VS Code with Live Server
- Write and test your very first JavaScript program

01 What is JavaScript?

JavaScript (JS) is a lightweight programming language that runs inside your browser and makes web pages interactive. Every time you see a button animate, a form validate before submitting, or a product image change on hover on sites like Flipkart or Meesho — that behaviour is powered by JavaScript. Without JS, a webpage is just a static document, like a printed pamphlet. With JS, it becomes a living application.

CONCEPT — Client-Side vs Server-Side

Client-side code runs in the visitor's browser — no server trip needed. Server-side code (Node.js, PHP, Python) runs on a machine in a data centre and sends results back. Think of it this way: when you type your name into a registration form on a Gujarat government portal and the page immediately says 'Name cannot be empty' — that check happened client-side, right in your browser, before anything was sent to the server.

Client-Side JS

Runs in the browser. Instant feedback — no server round-trip. Controls what the user sees and interacts with. Examples: form validation, dropdowns, modals, animations.

Server-Side JS (Node.js)

Runs on a web server. Handles databases, authentication, file storage. The browser never sees this code directly. Examples: saving form data to MySQL, generating PDF receipts.

02 How Browsers Read JavaScript

When your browser loads an HTML page, it reads the file top to bottom, line by line. Whenever it meets a `<script>` tag, it pauses HTML parsing and runs the JavaScript inside. This is why placement of your `<script>` tag matters: putting it just before the closing `</body>` tag ensures the entire page structure is built before JS tries to interact with it. Think of it like a construction team — the walls must exist before the painter arrives.

<code><!DOCTYPE html></code>	Declares a modern HTML5 document
<code><html></code>	Root element of the page
<code><body></code>	Visible content goes here
<code><h1>My Bio Link</h1></code>	A heading on the page
<code><script src="app.js"></script></code>	Links external JS file — placed just before </body>
<code></body></code>	
<code></html></code>	

PRO TIP — Inline vs External JS

You can write JS directly inside a `<script>` tag (inline) or in a separate .js file linked via `src`. For any project with more than 10 lines of code, always use a separate file. It keeps your HTML clean and lets the browser cache your script — meaning faster load times for repeat visitors, which is critical when 70% of Indian web users are on mobile data.

03 Setting Up Your Environment

You need two free tools: VS Code (code editor) and the Live Server extension. Live Server automatically refreshes your browser every time you save a file — no manual F5 needed. Download VS Code from code.visualstudio.com, then open the Extensions panel (Ctrl+Shift+X) and search 'Live Server' by Ritwick Dey. Click Install, right-click your HTML file, and choose 'Open with Live Server'. Your page opens at <http://127.0.0.1:5500> — a local web server running entirely on your own machine.

TRY THIS LIVE

Open VS Code -> create a folder called bio-link -> inside it create index.html and app.js -> add a basic HTML skeleton -> in app.js type: `console.log('Hello from JS!')` -> open with Live Server -> press F12 in the browser -> click the Console tab. You should see your message. This is your first working JavaScript program.

04 The Browser Console — Your Best Friend

The browser's Developer Tools (F12 or right-click -> Inspect) contain a Console tab where JavaScript runs live. You can type any JS directly into the console and see results instantly — no file saving needed. Professional developers use the console every single day for testing ideas, debugging errors, and inspecting values. Every session in this module will use the console to verify your code before adding it to your project.

<code>console.log('Welcome to JS!')</code>	Prints a message in the console — most-used debugging tool
<code>alert('Welcome to your Bio-Link App!')</code>	Shows a pop-up dialog in the browser — use sparingly
<code>console.log(2 + 3)</code>	JS evaluates the expression and prints 5
<code>typeof 'Aarya'</code>	Returns 'string' — useful for checking variable types

CASE STUDY — Zepto (Mumbai)

Context — Zepto is a fast-commerce startup operating across Mumbai, Bengaluru, Ahmedabad, and other Indian cities, delivering groceries in 10 minutes.

Challenge — Product pages needed to update cart count, estimated delivery time, and stock availability instantly without reloading the page — crucial for a time-sensitive 10-minute delivery promise.

What they did — Zepto's frontend team built heavily client-side JavaScript logic that updates UI elements in real time as users add items. DOM manipulation and event listeners — the exact concepts in Sessions 6–7 of this module — power this experience.

Result — Users see live cart totals and delivery estimates without any page refresh, making the checkout experience feel as fast as the delivery itself.

Takeaway — What feels like magic UX is JavaScript running in the browser — the same environment you just set up today.

Source: Zepto Engineering Blog, 2023; publicly documented tech stack

JS DEVELOPER DEMAND — INDIA

Rs 6–14 LPA

Average salary range for a JavaScript / React developer in Ahmedabad & Surat (Naukri.com, 2024)

*“JavaScript is the only language that runs natively in every browser on earth.
Learning it is not optional for a web developer — it is the foundation.”*

WHERE TO GO NEXT

You now know what JS is, how the browser runs it, and how to see output in the console. Every session from here builds on this foundation.

Explore deeper — MDN Web Docs: 'What is JavaScript?' —
developer.mozilla.org/en-US/docs/Learn/JavaScript/First_steps/What_is_JavaScript

Try on your own — Open any website you use (Swiggy, Naukri, Amazon.in) -> press F12 -> go to Console -> type `document.title` and press Enter. See what JS knows about the page.

Connect the dots — Session 02 introduces variables — the containers JS uses to store all the profile data in your Bio-Link App (name, follower count, bio text).

Going further — Read about the V8 JavaScript Engine (used in Chrome and Node.js) to understand how JS gets compiled and optimised at runtime — search 'V8 engine explained simply'.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript teacher for beginners. Explain in simple Hinglish what the difference is between client-side and server-side JavaScript. Use an example of a Flipkart product page. Keep it under 150 words.

SESSION

02

60 MIN · FUNDAMENTALS

Variables & Data Types

◆ Learning Objectives

- Declare variables using var, let, and const and explain the difference between them
- Understand scope — global, function, and block — and why it matters
- Identify all seven primitive data types in JavaScript
- Distinguish between primitive and reference types in memory
- Explain dynamic typing and why JS does not require type declarations
- Represent real Bio-Link App profile data using correctly typed variables

01 Why Variables Exist

A variable is a named container that stores a value in memory so your program can use and update it later. Without variables, every value would be hard-coded — you could never build an app that shows different content for different users. On a platform like Naukri.com, a single page template uses variables to show your name, your city, your skills, and your salary expectation — all different for every visitor.

02 var, let, and const — Which to Use?

JavaScript has three keywords for declaring variables. var is the original keyword from 1995 — it has confusing scoping rules and should be avoided in modern code. let was introduced in ES6 (2015) and is used for values that will change. const is also from ES6 and is used for values that should never be reassigned. A simple rule: start with const for everything; switch to let only when you know the value must change.

```
const userName = 'Aarya';
let followers = 1500;
let isVerified = false;
const appName = 'Bio-Link';
followers = followers + 100;
userName = 'Rohan';
```

Profile name — will never change during this session, so const is correct

Follower count — can grow, so let is correct

Boolean flag — will flip to true once verified

App name is fixed — const is appropriate

Reassigning let works fine — followers is now 1600

ERROR: Cannot reassign a const — JS throws TypeError

CONCEPT — Scope

Scope defines where in your code a variable can be accessed. var is function-scoped — it leaks out of if-blocks and loops, causing hard-to-find bugs. let and const are block-scoped — they stay inside the { } where they were created. Imagine a register at your TOPS batch: names written inside the batch register (block) are not visible outside. That is block scope.

03 Primitive Data Types

JavaScript has seven primitive types — these are the basic building blocks of all data. A primitive value is stored directly in memory and is immutable, meaning JS creates a new value instead of changing the original. Knowing the correct type for each piece of data prevents subtle bugs, especially when comparing values or sending data to an API.

Type	Example	Bio-Link App Use
String	'Aarya Patel'	User name, bio text, link URLs
Number	1500, 4.8	Follower count, average rating
Boolean	true / false	isVerified, isPublic, isDarkMode
Undefined	let city;	Variable declared but not yet assigned
Null	let deletedLink = null;	Intentionally empty — link was removed
BigInt	9007199254740991n	Very large integers (rare in UI work)
Symbol	Symbol('id')	Unique identifiers (advanced — covered later)

04 Reference Types & Dynamic Typing

CONCEPT — Primitive vs Reference

Primitives (string, number, boolean) are copied by value — two variables each hold their own copy. Objects and Arrays are copied by reference — two variables point to the same place in memory. Change the object through one variable and the other variable sees the change too. This surprises most beginners. Think of it like sharing a Google Doc link: editing the doc affects everyone who has the link.

```
let a = 10; let b = a; b = 20;
console.log(a);
let profile1 = { name: 'Aarya' };
let profile2 = profile1;
profile2.name = 'Rohan';
console.log(profile1.name);
```

Primitive copy — a is still 10, b is 20 (independent)
Prints 10 — unchanged
Object stored as a reference in memory
Both variables point to the SAME object
Mutates the shared object
Prints 'Rohan' — profile1 was also changed!

JavaScript is dynamically typed — you do not declare the type of a variable. JS infers it from the value assigned. You can even reassign a variable from one type to another: let x = 42; then x = 'hello'; is perfectly valid. This flexibility

speeds up development but requires discipline — always know what type your variable holds, especially before sending it to an API or comparing it with `===`.

TRY THIS LIVE

Open your browser console (F12). Type: `let followers = 1500`; then `typeof followers` — you get `'number'`. Now type: `followers = '1500'`; then `typeof followers` — you get `'string'`. Same variable name, different type. Notice how JS allowed this without any error. Now try `followers + 100` — you get `'1500100'` (string concatenation), not 1600. This is a classic dynamic-typing bug.

PRO TIP — Always Use `===` Not `==`

JavaScript has two equality operators: `==` (loose) and `===` (strict). Loose equality does type coercion — it converts types before comparing, leading to surprises: `0 == false` is true, `"" == false` is true. Strict equality never converts types: `0 === false` is false. Always use `===` in your code. Many senior developers consider `==` a JavaScript design mistake.

TYPE-RELATED BUGS IN JS CODEBASES

~22%

Proportion of JS bugs linked to unexpected type coercion or wrong type assumptions (Airbnb Engineering analysis, 2019)

WHERE TO GO NEXT

You can now store any kind of data in variables and understand how JS manages types. This is the data layer of every app you will build.

Explore deeper — MDN Docs: 'JavaScript data types and data structures' — developer.mozilla.org/en-US/docs/Web/JavaScript/Data_structures

Try on your own — In your browser console, declare five variables that represent a social media profile (name, bio, follower count, is verified, profile URL). Check the type of each using `typeof`.

Connect the dots — Session 03 uses these variables inside operators and conditions — you will write logic that checks follower count to decide whether a user gets a Verified badge.

Going further — Look up 'TypeScript' — it is JavaScript with strict types enforced at compile time. After this course, learning TypeScript will make you significantly more hireable.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript tutor. Explain the difference between `let`, `const`, and `var` in simple Hinglish with a real-life Indian example. Also explain what 'scope' means using a school classroom analogy. Keep it under 150 words.

SESSION

03

60 MIN · FUNDAMENTALS

Operators & Conditions

◆ Learning Objectives

- Use arithmetic, assignment, comparison, and logical operators correctly
- Explain truthy and falsy values and predict how JS evaluates them
- Write if / else if / else chains to control program flow
- Use the ternary operator as a concise alternative to if/else
- Apply the nullish coalescing (??) and optional chaining (?.) operators
- Build a Verified Creator badge logic using real follower-count conditions

01 Arithmetic & Assignment Operators

Operators are symbols that perform operations on values. Arithmetic operators do math — addition, subtraction, multiplication, division, and the modulo operator (%) which returns the remainder of a division. Assignment operators store results back into variables. These are the building blocks of all calculation logic in your app — from computing total followers to calculating a discount price on an e-commerce site like Ajio or Myntra.

```
let followers = 1500;
let newFollowers = followers + 200;
followers += 200;

let clicks = 340; let links = 4;
let avgClicks = clicks / links;
let isEvenFollowers = followers % 2 === 0;
```

Starting follower count

Arithmetic: addition — result is 1700

Assignment shorthand — same as followers = followers + 200

Declare two numbers

Division — result is 85 avg clicks per link

Modulo — checks if follower count is even

02 Comparison & Logical Operators

Comparison operators evaluate a relationship between two values and always return a boolean — true or false. They are the questions your program asks before making a decision. Logical operators combine multiple boolean conditions: && (AND) requires all conditions to be true, || (OR) requires at least one, and ! (NOT) flips a boolean. On Instagram, the verified badge logic checks: are followers above a threshold AND is the account authentic AND is the profile complete? All three must be true.

Operator	Meaning	Example (followers = 1500)
===	Strictly equal	followers === 1500 -> true
!==	Strictly not equal	followers !== 1000 -> true

Operator	Meaning	Example (followers = 1500)
>	Greater than	followers > 1000 -> true
>=	Greater than or equal	followers >= 1500 -> true
&&	AND — both must be true	followers > 1000 && isVerified -> false
	OR — at least one true	followers > 5000 isVerified -> false
!	NOT — flips boolean	!isVerified -> true

03 Truthy & Falsy Values

CONCEPT — Truthy & Falsy

In JavaScript, every value has an inherent boolean meaning when used in a condition. Six values are falsy — they behave like false: 0, "" (empty string), null, undefined, NaN, and false itself. Everything else is truthy. This matters enormously in real code: if (userName) will be false if userName is an empty string or undefined. Think of it like a food delivery app — if the delivery address field is empty, the order button should be disabled. JS conditions check truthiness, not just exact boolean values.

```
console.log(Boolean(0));           false — zero is falsy
console.log(Boolean(''));         false — empty string is falsy
console.log(Boolean(null));        false — null is falsy
console.log(Boolean('Aarya'));     true — any non-empty string is truthy
console.log(Boolean(1500));         true — any non-zero number is truthy
console.log(Boolean([]));          true — even an empty array is truthy (surprises beginners!)
```

04 if / else if / else — Decision Making

Conditional statements let your program take different paths depending on data. An if block runs when its condition is truthy. An else if adds a second check. An else is the fallback when no condition matched. You can chain as many else if blocks as you need. Real apps use these constantly: show a 'Premium' badge if the user has a paid plan, show a 'New' badge if they joined this month, otherwise show nothing.

```
let followers = 1500;           Follower count from the profile
if (followers > 100000) {        Mega influencer — 1 lakh+ followers
  console.log(' Mega Creator');
} else if (followers > 10000) {   Macro influencer — 10k–1 lakh
```

```

    console.log(' Verified Creator');
  } else if (followers > 1000) {
    console.log('■ Rising Creator');
  } else {
    console.log('Keep growing!');
  }

```

Micro influencer — 1k–10k

Everyone else

With followers = 1500, output is: ■ Rising Creator

05 Ternary Operator & Modern Shortcuts

CONCEPT — Ternary Operator

The ternary operator is a one-line if/else: `condition ? valueIfTrue : valueIfFalse`. It is called 'ternary' because it takes three operands. Use it for simple yes/no decisions, especially when assigning a value. Avoid nesting ternaries — if your condition logic is complex, use a regular if/else for readability.

```

let badge = followers > 1000 ? ' Verified Creator' : 'Keep growing!';

```

Ternary assigns one of two strings based on the condition

```

console.log(badge);
let bio = userName ?? 'Anonymous';
let city = profile?.location?.city;

```

Prints: Verified Creator (since 1500 > 1000)

?? (nullish coalescing) — uses 'Anonymous' only if userName is null or undefined

?. (optional chaining) — returns undefined safely if profile or location doesn't exist

TRY THIS LIVE

In your browser console, declare: `let followers = 850; let isVerified = false;` Now write a ternary that returns 'Eligible for badge' if `followers > 1000` OR `isVerified` is true, else returns 'Not eligible yet'. Test it by changing the values of `followers` and `isVerified`. This is the exact logic Instagram and YouTube use for creator tier labels.

PRO TIP — switch vs if/else

When you are comparing one variable against many fixed values (like checking a user's subscription plan: 'free', 'basic', 'pro', 'enterprise'), a switch statement is cleaner than a long if/else chain. However, switch uses `==` not `===`, and forgetting a break causes fall-through bugs. For most modern React code, you will use object maps or ternaries more often than switch.

WHERE TO GO NEXT

You can now make decisions in code based on data. Every interactive feature in your app — show/hide, enable/disable, badge or no badge — flows from these concepts.

Explore deeper — MDN Docs: 'Expressions and operators' —
developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Expressions_and_Operators

Try on your own — Build a simple price-discount calculator in the console. If price > 1000, apply 20% discount. If price > 500, apply 10%. Otherwise no discount. Test with three different prices.

Connect the dots — Session 04 introduces loops — instead of writing conditions once for one item, you will apply them to every item in a list, which is how the Bio-Link App will display all social links.

Going further — Research 'short-circuit evaluation' in JavaScript — how && and || can be used to conditionally execute code in a single line, a pattern used heavily in React JSX.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript mentor. Explain the concept of truthy and falsy values in Hinglish with three funny Indian examples. Then show a simple ternary operator example using a cricket match score condition. Keep it under 150 words.

SESSION

04

60 MIN · FUNDAMENTALS

Loops & Iteration

◆ Learning Objectives

- Write for, while, and do-while loops and explain when to use each
- Iterate over arrays using for-of and forEach
- Use break and continue to control loop execution
- Avoid infinite loops by ensuring loop conditions eventually become false
- Loop through an array of social links and print each to the console
- Understand how iteration connects to dynamic rendering in React

01 Why Loops Exist

A loop repeats a block of code multiple times without you writing it again and again. Without loops, displaying 10 social links would mean writing 10 separate `console.log` statements. With a loop, you write one statement and let JS do the repetition. Every list you see on a web app — job listings on Naukri, product cards on Amazon.in, notifications on WhatsApp — is rendered using a loop over an array of data.

02 The for Loop — Most Common Loop

The for loop is the most commonly used loop in JavaScript. It has three parts in its header: an initialiser (run once at start), a condition (checked before each iteration), and an increment (run after each iteration). The loop runs as long as the condition is truthy. Use a for loop when you know exactly how many times you need to repeat — like iterating over an array using its index.

```
for (let i = 0; i < 5; i++) {  
    console.log('Link number: ' + i);  
}  
const links = ['Instagram', 'YouTube', 'LinkedIn', 'Twitter'];  
    Array of social links  
for (let i = 0; i < links.length; i++) {  
    console.log(links[i]);  
}
```

i starts at 0, loop runs while i < 5, i increases by 1 each time
Prints Link number: 0, 1, 2, 3, 4
Loop ends when i reaches 5
Loop through every index in the array
Prints each link title by index

03 while & do-while Loops

CONCEPT — while vs do-while

A while loop checks its condition BEFORE executing the block. If the condition is false from the start, the block never runs. A do-while loop executes the block ONCE first, then checks the condition — guaranteeing at least one execution. Use while when repetition is optional; use do-while when the first execution must always happen. Think of a school attendance roll-call: the teacher calls names while students are still present (while). But on the first day, the teacher always takes attendance at least once even before knowing who enrolled (do-while).

```
let followers = 500;
while (followers < 1000) {
  followers += 100;
  console.log('Followers: ' + followers);
}
```

Starting follower count

Keep looping until follower goal is reached

Simulate gaining 100 followers per iteration

Stops when followers reaches 1000 — prints 600, 700, 800, 900, 1000

```
let attempts = 0;
do {
  attempts++;
  console.log('Attempt: ' + attempts);
} while (attempts < 3);
```

Block runs at least once regardless of condition

Increment attempt counter

Continues while attempts < 3 — prints 1, 2, 3

04 for-of and forEach — Modern Array Iteration

ES6 introduced cleaner ways to loop through arrays. for-of gives you the value of each element directly — no index management needed. forEach is an array method that accepts a callback function and calls it once for each element. Both are cleaner and less error-prone than a traditional for loop for simple iteration. In React, you will almost always use .map() (covered in Session 16) which builds on this same idea.

```
const links = [{title:'Instagram', url:'https://instagram.com/aarya'},
{title:'YouTube', url:'https://youtube.com/@aarya'}];
```

Array of link objects

```
for (const link of links) {
  console.log(link.title + ' -> ' + link.url);
}
```

for-of: link holds each object directly — no index

Access object properties directly

```
links.forEach(function(link) {
  console.log(link.title);
});
```

forEach: classic callback syntax

Same output — prints Instagram, YouTube

```
links.forEach(link => console.log(link.title));
```

Arrow function shorthand — same result, cleaner

TRY THIS LIVE

In your browser console, create an array with at least 4 social link objects (each with title, url, and clicks). Use a for-of loop to print each link in this format: 'Instagram -> https://... -> 234 clicks'. Then add a condition inside the loop: if clicks > 200, also print 'HIGH TRAFFIC'. This mini-challenge combines loops with conditionals from Session 03.

05 break, continue & Avoiding Infinite Loops

CONCEPT — break & continue

break immediately exits the loop — all remaining iterations are cancelled. continue skips the rest of the current iteration and jumps to the next one. These give you fine-grained control over loop flow. On an e-commerce site like Meesho, a search loop might break as soon as it finds an exact product match, saving processing time on large catalogues.

```
for (let i = 0; i < 10; i++) {
  if (i === 5) break;
  console.log(i);
}
for (let i = 0; i < 6; i++) {
  if (i === 3) continue;
  console.log(i);
}
```

Stops the entire loop when i hits 5 — only prints 0 to 4

Skips index 3 — prints 0, 1, 2, 4, 5

WARNING — Infinite Loops

An infinite loop runs forever because its condition never becomes false, freezing your browser tab. The most common cause is forgetting to increment the counter: while (i < 10) { console.log(i); } — i never changes, loop never ends. Always double-check that your loop variable moves toward the exit condition with every iteration. If your browser freezes during practice, close the tab immediately.

LINEs OF LOOP CODE IN A TYPICAL REACT APP

~40%

Proportion of React rendering code that uses array iteration methods (map, forEach, filter) — highlights how central looping is to front-end development

WHERE TO GO NEXT

You can now repeat operations across any collection of data. This is how your Bio-Link App will eventually render every social link card from an array automatically.

Explore deeper — MDN Docs: 'Loops and iteration' —
developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Loops_and_iteration

Try on your own — Create an array of 5 Gujarati cities (Ahmedabad, Surat, Vadodara, Rajkot, Gandhinagar). Loop through them and print each with its index number. Then use continue to skip Gandhinagar.

Connect the dots — Session 05 introduces functions — you will wrap your loop logic inside reusable functions so the same display code can be called for any dataset, not just hard-coded links.

Going further — Look up 'generator functions' in JavaScript — a powerful ES6 feature that lets you pause and resume loops, useful for handling paginated API responses.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript teacher. Explain the difference between for, while, and for-of loops in simple Hinglish. Use an example of a conductor on an Ahmedabad BRTS bus checking tickets. Keep it under 150 words.

SESSION

05

60 MIN · FUNDAMENTALS

Functions & Reusability

◆ Learning Objectives

- Declare functions using function declarations, expressions, and arrow functions
- Pass parameters into functions and return values from them
- Explain the difference between a function definition and a function call
- Understand hoisting and how it affects function declarations vs expressions
- Write pure, single-purpose functions that are easy to test and reuse
- Build a `createButton()` factory function for the Bio-Link App

01 What is a Function and Why Use One?

A function is a named, reusable block of code that performs a specific task. Instead of writing the same logic multiple times, you define it once and call it wherever needed. Functions are the core unit of code organisation in JavaScript — every React component you will write in Module 2 is essentially a function that returns HTML. Think of a function like a Zomato delivery workflow: the steps (accept order -> pick up -> navigate -> deliver -> confirm) are defined once and executed every time a new order arrives, for every order, without rewriting the steps.

02 Three Ways to Define a Function

CONCEPT — Declaration vs Expression vs Arrow

Function declarations are hoisted — JS moves them to the top of their scope, so you can call them before they appear in the file. Function expressions and arrow functions are NOT hoisted — you must define them before calling them. Arrow functions also do not have their own 'this' keyword, which becomes important in React event handlers. For most modern JS and all React code, arrow functions are the preferred style.

// 1. Function Declaration (hoisted)

```
function greetUser(name) {
```

```
  return 'Welcome, ' + name + '!';
```

```
}
```

// 2. Function Expression (not hoisted)

```
const showBadge = function(followers) {
```

```
  return followers > 1000 ? ' Verified' : '■ Rising';
```

```
};
```

// 3. Arrow Function (preferred in modern JS)

Classic declaration — can be called before this line in the file

return sends a value back to the caller

Assigned to a variable — cannot be called before this line

```
const calcEngagement = (likes, followers) => (likes / followers) * 100;
```

Implicit return when body is a single expression

03 Parameters, Arguments & Return Values

Parameters are the named placeholders in the function definition. Arguments are the actual values passed when calling the function. A function can have zero or many parameters. The return keyword sends a value back to wherever the function was called — without return, a function returns undefined. A function should do one thing well. If a function is doing three unrelated things, it should be three separate functions — this is called the Single Responsibility Principle and it makes code dramatically easier to maintain.

```
const createButton = (label, color) => {
  const btn = document.createElement('button');
  btn.textContent = label;
  btn.style.backgroundColor = color;
  return btn;
};
const instagramBtn = createButton('Instagram', '#E1306C');
const youtubeBtn = createButton('YouTube', '#FF0000');
document.body.appendChild(instagramBtn);
```

label and color are parameters

Creates a new HTML button element

Sets the button text

Sets background colour

Returns the button element so the caller can use it

Call with two arguments — reuse for any social link

Same function, different arguments — zero duplication

Adds the button to the page

TRY THIS LIVE

Open your browser console. Define an arrow function called formatLink that takes a title and url as parameters and returns a formatted string: 'Instagram -> https://instagram.com/aarya'. Call it three times with different social media platforms. Notice how you write the formatting logic once and reuse it for every link — this is the core power of functions.

04 Default Parameters & Rest Parameters

```
const greet = (name = 'Guest', role = 'Visitor') => {
  return `Welcome, ${role} ${name}!`;
};
greet();
```

Default params used if caller doesn't pass those arguments

Returns: 'Welcome, Visitor Guest!' — both defaults used

```
greet('Aarya', 'Creator');
```

Returns: 'Welcome, Creator Aarya!' — defaults overridden

```
const sumAll = (...numbers) => numbers.reduce((acc, n) => acc + n, 0);
```

...numbers (rest param) collects any number of arguments into an array

```
sumAll(10, 20, 30, 40);
```

Returns 100 — works with any count of arguments

05 Higher-Order Functions — Functions Using Functions

CONCEPT — Higher-Order Functions

A higher-order function is a function that accepts another function as an argument, or returns a function as its result. This is one of the most powerful patterns in JavaScript and the foundation of how array methods like `forEach`, `map`, `filter`, and `reduce` work. In React, you will pass functions as props to child components every day — that is higher-order thinking in action.

```
const applyToAll = (items, transformFn) => items.map(transformFn);
```

Higher-order function — accepts an array and a function

```
const links = ['instagram', 'youtube', 'linkedin'];
```

```
const upperLinks = applyToAll(links, link => link.toUpperCase());
```

Passes an arrow function as the second argument

```
console.log(upperLinks);
```

Prints: ['INSTAGRAM', 'YOUTUBE', 'LINKEDIN']

PRO TIP — Keep Functions Pure

A pure function always returns the same output for the same input and has no side effects (it does not change anything outside itself). Pure functions are easy to test, easy to debug, and safe to reuse. When you start writing React, pure functions will be crucial — React's rendering system assumes components behave predictably like pure functions.

CASE STUDY — Juspay (Bengaluru)

Context — Juspay is India's largest payments infrastructure company, processing transactions for Swiggy, Ola, Dunzo, and hundreds of other apps.

Challenge — Their checkout SDK needed to render different payment UI widgets (UPI, card, netbanking, EMI) depending on user context — without duplicating the rendering logic for each payment type.

What they did — Engineers built factory functions and higher-order functions that accepted a payment type and returned the correct UI builder. One function definition handled all widget types by parameterising the differences.

Result — The SDK reduced UI code duplication by over 60%, making it faster to add new payment methods

and easier to fix bugs across all widgets simultaneously.

Takeaway — Functions with parameters eliminate duplication — the same principle applies to your `createButton()` today and your React components tomorrow.

Source: Juspay Engineering Blog, 2022

WHERE TO GO NEXT

Functions are the backbone of every JavaScript application. Everything from here — DOM manipulation, events, async code, React components — is built on the function concepts you practised today.

Explore deeper — MDN Docs: 'Functions — reusable blocks of code' — developer.mozilla.org/en-US/docs/Learn/JavaScript/Building_blocks/Functions

Try on your own — Write three functions for the Bio-Link App: `addLink(title, url)` that logs a formatted link, `removeLink(title)` that logs a removal message, and `countLinks(linksArray)` that returns the total count. Call each function from the console.

Connect the dots — Session 06 applies functions to the DOM — your `createButton()` function will actually add real buttons to the live page, not just log things in the console.

Going further — Research 'closures' in JavaScript — a function that remembers the variables from its outer scope even after that outer function has finished running. Closures power many React hooks including `useState`.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript instructor. Explain the concept of higher-order functions in simple Hinglish. Use an analogy of a catering manager in Ahmedabad who gives different chefs (functions) different tasks. Show a short code example. Keep it under 150 words.

SESSION

06

60 MIN · BROWSER & DOM

DOM Manipulation – Selecting Elements

◆ Learning Objectives

- Explain what the DOM is and how the browser builds it from HTML
- Select elements using `getElementById`, `querySelector`, and `querySelectorAll`
- Read and update element content using `innerHTML`, `textContent`, and `value`
- Change element styles dynamically using the `style` property and `classList`
- Create new elements with `createElement` and add them to the page with `appendChild`
- Dynamically add a new social link card to the Bio-Link App from an input form

01 What is the DOM?

The DOM (Document Object Model) is the browser's live, in-memory representation of your HTML page as a tree of objects. When the browser loads your HTML, it parses each tag and creates a corresponding JavaScript object — called a node — for it. These nodes are connected in a parent-child tree structure. JavaScript can reach into this tree, read any node's properties, change them, add new nodes, or delete existing ones — and the browser instantly updates the visual page to reflect every change.

CONCEPT — DOM Tree Structure

Your HTML file is the blueprint. The DOM is the building the browser constructs from that blueprint. The document object is the root — it represents the entire page. Under it sits the `html` node, then `head` and `body`, then all the tags inside `body`. When you write `document.getElementById('title')`, you are navigating this tree to find a specific node, just like searching for a specific room in a building using its room number.

02 Selecting Elements — `querySelector` & Friends

Before JavaScript can do anything with an HTML element, it must first find (select) it. JavaScript offers several selector methods. `getElementById` is the fastest — it finds one element by its unique `id` attribute. `querySelector` accepts any CSS selector string and returns the first matching element. `querySelectorAll` returns a `NodeList` (similar to an array) of all matching elements. In modern codebases, `querySelector` and `querySelectorAll` are preferred because they use familiar CSS syntax.

```
const title = document.getElementById('profile-title');
```

Finds the one element with `id='profile-title'`

```
const bio = document.querySelector('.bio-text');
```

Finds the FIRST element with class `'bio-text'`

```
const allCards = document.querySelectorAll('.link-card');
  Returns ALL elements with class 'link-card' as a NodeList
const firstCard = document.querySelector('.link-card');
  Only the first card — same as allCards[0]
allCards.forEach(card => console.log(card.textContent));
  Iterate NodeList with forEach — prints each card's text
```

03 Reading & Updating Content

Once you have selected an element, you can read or modify its content using three key properties. `textContent` gets or sets the raw text inside an element — it ignores HTML tags. `innerHTML` gets or sets the HTML markup inside an element — it parses tags. `value` reads the current content of form inputs (text fields, dropdowns). Use `textContent` by default; use `innerHTML` only when you deliberately need to inject HTML markup, as it can create security vulnerabilities if used with untrusted user input.

```
const nameEl = document.getElementById('profile-name');
  Select the name element
console.log(nameEl.textContent);
  Read current text — prints e.g. 'Aarya Patel'
nameEl.textContent = 'Aarya Shah';
  Update text — page updates instantly
nameEl.innerHTML = '<strong>Aarya Shah</strong>';
  Inject HTML — name appears bold
const input = document.querySelector('#link-title-input');
  Select the text input
const enteredTitle = input.value;
  Read whatever the user typed into the input
input.value = '';
  Clear the input field after reading
```

04 Changing Styles with `style` & `classList`

CONCEPT — `style` vs `classList`

You can change an element's appearance two ways. The `style` property lets you set individual CSS properties directly in JavaScript (`element.style.backgroundColor = 'red'`). `classList` lets you add, remove, or toggle CSS class names — which is cleaner because your styles stay in the CSS file and your JS only manages class names. Always prefer `classList` for toggling themes, hover states, and show/hide functionality. Reserve `style` for dynamic values that genuinely can't live in a CSS file — like a progress bar width calculated from data.

```
const card = document.querySelector('.link-card');
  Select a card
card.style.backgroundColor = '#E1306C';
  Direct style — Instagram pink
```

```
card.style.borderRadius = '12px';
card.classList.add('active');

card.classList.remove('hidden');
card.classList.toggle('dark-mode');

card.classList.contains('active');
```

Direct style — rounded corners

Adds the 'active' class — applies CSS you wrote for .active

Removes 'hidden' class — makes the card visible

Adds class if absent, removes if present — one-line theme toggle

Returns true/false — check if class is currently applied

05 Creating & Inserting New Elements

You can create brand-new HTML elements in JavaScript using `document.createElement()`, configure them, and then insert them into the DOM. This is how dynamic UIs work — the HTML file does not contain the final list of social links; JavaScript creates each link card on the fly from data. This is also exactly how React renders components: instead of hand-writing HTML, it calls functions that create and insert DOM nodes based on your data.

```
function createLinkCard(title, url) {
  const card = document.createElement('div');
  card.className = 'link-card';
  card.innerHTML = `<a href='${url}'>${title}</a>`;
  Inject an anchor tag inside the div
  return card;
}
```

Factory function — creates one link card from data

Create a new <div> element

Add CSS class for styling

Return the configured element

```
const container = document.querySelector('#links-container');
Select the parent container on the page
container.appendChild(createLinkCard('Instagram', 'https://instagram.com/aarya'));
Insert the new card into the DOM — appears on page immediately
```

TRY THIS LIVE

In your `index.html`, add a `<div id='links-container'></div>` and a form with two inputs (title, url) and a button. In `app.js`, select the inputs and button. On button click (we'll cover events properly next session — for now use `onclick` on the button tag), call `createLinkCard()` with the input values and append the result to `#links-container`. Then clear both inputs. You should see a new link card appear on the page each time you submit.

PRO TIP — innerHTML Security Warning

Never inject user-supplied text directly using `innerHTML` — it enables Cross-Site Scripting (XSS) attacks. If a malicious user types `<script>stealCookies()</script>` into your input and you insert it with `innerHTML`, that script executes. Always use `textContent` for user-provided text, or sanitise input before injecting HTML. This is a real security vulnerability that has affected major Indian banking portals.

WHERE TO GO NEXT

You can now reach into any part of your page, read it, change it, and add to it. Your Bio-Link App now dynamically creates cards from input — the core mechanic of any interactive web app.

Explore deeper — MDN Docs: 'Introduction to the DOM' —
developer.mozilla.org/en-US/docs/Web/API/Document_Object_Model/Introduction

Try on your own — Add a 'Remove' button inside each link card that, when clicked, removes that specific card from the DOM using `element.remove()`. Combine this with what you already know about `createLinkCard`.

Connect the dots — Session 07 replaces the basic `onclick` attribute with proper `addEventListener` — giving you full control over every user interaction, not just button clicks.

Going further — Look up `document.createDocumentFragment()` — a performance optimisation technique for adding many elements to the DOM at once without causing multiple re-renders, used in high-performance UI libraries.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a web development tutor. Explain in Hinglish what the DOM is and why JavaScript needs it. Use the analogy of a government office file system in Gujarat where each file is a DOM node. Keep it under 150 words.

SESSION

07

60 MIN · BROWSER & DOM

DOM Events & Interactivity

◆ Learning Objectives

- Explain what a DOM event is and how the browser fires events
- Attach event handlers using onclick attributes, the on-property, and addEventListener
- Explain why addEventListener is preferred over the other two approaches
- Use the event object to read event details like target, key pressed, and mouse position
- Handle click, input, change, keydown, and submit events
- Wire the Bio-Link App's Add Link button to dynamically create and display link cards

01 What is a DOM Event?

An event is a signal that something happened in the browser — a user clicked a button, typed a character, moved the mouse, submitted a form, or the page finished loading. The browser fires these events constantly. Your job as a JavaScript developer is to listen for the specific events you care about and define what should happen in response. On every e-commerce site — Amazon, Flipkart, Myntra — every button click, filter selection, and search keystroke is handled by a JavaScript event listener.

02 Three Ways to Handle Events

CONCEPT — onclick vs on-property vs addEventListener

Method 1 (HTML attribute): `<button onclick='myFn()>` — mixes HTML and JS, hard to maintain. Method 2 (DOM property): `element.onclick = myFn` — only one handler per event (assigning a second overwrites the first). Method 3 (addEventListener): `element.addEventListener('click', myFn)` — preferred. Multiple handlers can be attached, handlers can be removed, and event options (like once or capture) can be configured. Always use addEventListener in production code.

```
// Method 1 – Avoid (HTML attribute)
<button onclick="addLink()">Add Link</button>    Mixes logic into HTML — hard to manage at scale

// Method 2 – Avoid for multiple listeners
const btn = document.querySelector('#add-btn');
btn.onclick = addLink;                          Second assignment would overwrite this

// Method 3 – Preferred
btn.addEventListener('click', addLink);          Multiple listeners can be added; removable later
btn.addEventListener('click', logAnalytics);      Second listener on same event — both fire
```

03 The Event Object

Every event handler receives an event object as its first argument. This object contains rich information about what happened: which element was clicked (`event.target`), the mouse position (`event.clientX`, `event.clientY`), which key was pressed (`event.key`), and much more. Calling `event.preventDefault()` stops the browser's default behaviour — for example, preventing a form from reloading the page on submit. This is one of the most important patterns in all of web development.

```
const form = document.querySelector('#link-form');
  Select the form element
form.addEventListener('submit', function(event) {
  Listen for form submission
  event.preventDefault();           CRITICAL — stops browser from reloading the page
  const title = document.querySelector('#title-input').value;
  Read the title input
  const url = document.querySelector('#url-input').value;
  Read the URL input
  console.log('Adding link:', title, url);    Use the data — will call createLinkCard() next
});
```

04 Common Event Types

Event	Fires When	Bio-Link App Use Case
click	User clicks an element	Add Link button, Delete button, theme toggle
submit	Form is submitted (button or Enter)	Submit new social link form
input	Input value changes (every keystroke)	Live character count in bio field
change	Input loses focus with a new value	Dropdown selecting link category
keydown	Any key is pressed down	Keyboard shortcut to open add-link form
mouseover	Mouse enters an element's area	Show edit icon when hovering over a card
DOMContentLoaded	Page HTML is fully parsed (not images)	Initialise app and load saved links

TRY THIS LIVE

Wire up the full Add Link flow: select your form, add a submit listener with `preventDefault()`, read the two input values (title and url), call your `createLinkCard()` function from Session 06 with those values, append the result to `#links-container`, and then clear both inputs. Test: fill in the form, click Add Link — a new card should appear instantly. Repeat for multiple links. This is your first fully interactive feature.

05 Intro to Event Flow — Bubbling

CONCEPT — Event Bubbling

When an event fires on an element, it does not stop there — it bubbles up through every ancestor in the DOM tree, firing at each one. Click a button inside a div inside a section — all three elements receive the click event in order (button -> div -> section). This is called bubbling. You can stop it with `event.stopPropagation()`, but rarely need to. In Session 17 you will learn event delegation, which deliberately exploits bubbling to handle events on many child elements with a single parent listener.

```
document.querySelector('#links-container').addEventListener('click', function(e) {
  // One listener on the parent container
  console.log('Clicked element:', e.target.tagName);
  // e.target = the specific element that was clicked
  console.log('Listener on:', e.currentTarget.id);
  // e.currentTarget = element the listener is attached to
});
```

Even clicks on child cards bubble up and are caught here

PRO TIP — `removeEventListener` to Prevent Memory Leaks

Every `addEventListener` call creates a reference between the DOM element and your function, keeping both in memory. If you remove an element from the DOM without removing its listeners first, the listener and its closure remain in memory — a memory leak. Always remove listeners when they are no longer needed using `element.removeEventListener('click', myFn)`. Note: the function reference must be identical — anonymous functions cannot be removed, which is another reason to name your handler functions.

CASE STUDY — IRCTC (Indian Railways)

Context — IRCTC's online ticket booking portal handles millions of daily users across India, many on slow 2G/3G connections.

Challenge — The seat selection grid had hundreds of seat buttons. Attaching individual click listeners to each button caused massive memory consumption and slow page load — critical on low-end devices common in smaller Indian cities.

What they did — Frontend engineers refactored the seat selection to use a single delegated event listener on the parent grid container. One listener caught all seat clicks by reading `event.target` — reducing memory usage dramatically.

Result — Page interactivity improved measurably on low-memory devices, reducing booking abandonment rates among users on budget phones.

Takeaway — Event delegation — using bubbling with one parent listener — is not just cleaner code; it is a real performance optimisation that matters for India's device landscape.

Source: IRCTC frontend architecture discussions, Web Performance India community, 2023

WHERE TO GO NEXT

Your Bio-Link App is now interactive — users can add links and see them appear in real time. This is the foundation of every UI you will ever build.

Explore deeper — MDN Docs: 'Introduction to events' — developer.mozilla.org/en-US/docs/Learn/JavaScript/Building_blocks/Events

Try on your own — Add a Delete button inside each link card. When clicked, the button should remove that specific card. Hint: inside the `createLinkCard` function, create a delete button, add a click listener that calls `card.remove()`, and append it inside the card.

Connect the dots — Session 08 introduces arrays and objects — you will store your links as structured data instead of only as DOM nodes, making it possible to search, sort, and save them.

Going further — Research the 'capture phase' of event flow — the opposite direction to bubbling. `addEventListener` accepts a third argument `{capture: true}` that fires the listener on the way down the tree instead of up. Rarely needed but important in complex UI architectures.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript teacher for beginners. Explain event bubbling in simple Hinglish. Use the analogy of a complaint going from a student to a class teacher to a principal to the school board. Then show one line of code using `event.stopPropagation()`. Keep it under 150 words.

SESSION

08

60 MIN · DATA & STATE

Arrays & Objects – Data Structures

◆ Learning Objectives

- Create and access arrays using index notation
- Use core array methods: push, pop, shift, unshift, splice, slice, and indexOf
- Create object literals and access properties using dot and bracket notation
- Build arrays of objects — the most common data pattern in web development
- Loop through an array of objects to render structured content
- Model the Bio-Link App's link collection as an array of objects

01 Arrays — Ordered Collections

An array is an ordered list of values stored in a single variable. Each value is called an element; each element has a numeric index starting at 0. Arrays are the most common data structure in JavaScript — every product listing, notification feed, search result, or social link collection you interact with on the web is an array under the hood. On a platform like Swiggy, your order history, the restaurant list, and the items in your cart are all arrays of objects.

```
const platforms = ['Instagram', 'YouTube', 'LinkedIn', 'Twitter'];
```

Array of four strings

```
console.log(platforms[0]);
```

Index 0 -> 'Instagram' (arrays start at zero)

```
console.log(platforms[platforms.length - 1]);
```

Last element -> 'Twitter'

```
platforms.push('GitHub');
```

Adds 'GitHub' to the END — length becomes 5

```
platforms.pop();
```

Removes and returns the LAST element ('GitHub') — length back to 4

```
platforms.unshift('TikTok');
```

Adds 'TikTok' to the START — shifts all indices up by 1

```
platforms.shift();
```

Removes and returns the FIRST element ('TikTok')

02 splice & slice — Surgical Array Edits

CONCEPT — splice vs slice

These two methods sound similar but behave very differently. splice MODIFIES the original array — it can remove, replace, or insert elements at any position. slice creates a NEW array containing a copy of a portion of the original, without modifying it. Remember: spli-CE does not change, spli-CE = Copy Extract. Use slice when you want a safe copy; use splice when you deliberately need to mutate the array.

```
const links = ['Instagram', 'YouTube', 'LinkedIn', 'Twitter', 'GitHub'];
links.splice(2, 1);

console.log(links);
links.splice(1, 0, 'Pinterest');
const copy = links.slice(0, 3);

console.log(copy);
```

Removes 1 element starting at index 2 — removes 'LinkedIn'

['Instagram', 'YouTube', 'Twitter', 'GitHub']

Inserts 'Pinterest' at index 1, removes 0 elements

Returns NEW array with elements at index 0, 1, 2 — original unchanged

['Instagram', 'Pinterest', 'YouTube']

03 Objects — Structured Data Collections

An object is a collection of key-value pairs where each key (property) maps to a value. While an array is for an ordered list of similar things, an object is for a single entity with multiple named attributes. A social link has many properties — title, URL, icon colour, click count, whether it is active — and an object is the natural container for all of these. Objects are everywhere in web development: API responses, configuration settings, component props in React — all are objects.

```
const link = {
  title: 'Instagram',
  url: 'https://instagram.com/aarya',
  color: '#E1306C',
  clicks: 234,
  isActive: true
};
console.log(link.title);
console.log(link['url']);
```

Object literal syntax — curly braces contain key-value pairs

String property

String property

Hex colour for the card background

Number property

Boolean property — no trailing comma on last property

Dot notation — most common, use when key is a known identifier

Bracket notation — use when key is dynamic or has special characters

04 Arrays of Objects — The Most Common Pattern

In real applications, you almost never work with plain arrays of strings or numbers. You work with arrays of objects — each object representing one item with multiple attributes. This is exactly what a database table looks like when converted to JSON, and exactly what every API sends you. Your Bio-Link App's complete data model is an array of link objects — this is the single most important data structure you will use throughout your entire career as a JavaScript developer.

```
const links = [
  { title: 'Instagram', url: 'https://instagram.com/aarya', clicks: 234, isActive: true },
  { title: 'YouTube', url: 'https://youtube.com/@aarya', clicks: 512, isActive: true },
];
```

Array of link objects — the app's data model

```
{ title: 'LinkedIn', url: 'https://linkedin.com/in/aarya', clicks: 89, isActive: false
}
];
links.forEach(link => {                                Loop through every link object
  console.log(`${link.title}: ${link.clicks} clicks — ${link.isActive ? 'Active' :
  'Hidden'} `);
  Access properties and combine with ternary
});
```

TRY THIS LIVE

Declare a links array with at least 4 social link objects (title, url, clicks, isActive). Write a function addLink(title, url) that pushes a new link object into the array with clicks starting at 0 and isActive as true. Write a removeLink(title) that uses splice to remove the link matching the title. Write a displayLinks() function that loops through the array and logs each link. Call all three from the console and verify the array updates correctly.

PRO TIP — Never Mutate State Directly in React

In vanilla JS today, you are freely using push, pop, and splice to mutate arrays. When you reach React in Module 2, mutating state arrays directly breaks React's change detection. You will always create a new copy of the array when adding or removing items: [...existingLinks, newLink] instead of links.push(newLink). Build the habit now of thinking about the difference between mutating and copying — it will save you hours of React debugging later.

API RESPONSES THAT ARE ARRAYS OF OBJECTS

> 90%

Proportion of real-world REST API list responses (products, users, orders, posts) that return JSON arrays of objects — making this pattern the single most critical to master (REST API design surveys, 2023)

WHERE TO GO NEXT

You now have a proper in-memory data model for the Bio-Link App. All subsequent sessions will operate on this links array — reading, transforming, persisting, and rendering it.

Explore deeper — MDN Docs: 'Working with objects' —
developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Working_with_objects

Try on your own — Add a clicks property to each link object. Write a function getMostClicked() that loops through the links array and returns the link object with the highest clicks value. Log the winner.

Connect the dots — Session 09 saves this links array to localStorage so it persists across page reloads — the first step toward real data persistence before you learn about databases.

Going further — Look up the 'Spread operator for objects': `{...obj, key: newValue}` creates a new object with one property changed — the safe, React-friendly way to update object state without mutation.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript tutor. Explain the difference between an array and an object in simple Hinglish. Use an example of a train booking: the train has properties (name, number, departure city) — that is an object. A list of trains is an array of objects. Keep it under 150 words.

SESSION

09

60 MIN · DATA & STATE

JSON & LocalStorage (Persistence Basics)

◆ Learning Objectives

- Explain what JSON is and why it is the universal data format of the web
- Convert JavaScript objects to JSON strings using `JSON.stringify()`
- Parse JSON strings back into JavaScript objects using `JSON.parse()`
- Save, retrieve, update, and delete data in the browser's `localStorage`
- Persist the Bio-Link App's links array so it survives page reloads
- Recognise the limitations of `localStorage` compared to a real database

01 What is JSON?

JSON (JavaScript Object Notation) is a text format for representing structured data. It looks almost identical to a JavaScript object literal but with important differences: all keys must be in double quotes, and the only allowed value types are string, number, boolean, null, array, and object. No functions, no undefined, no Date objects. JSON is language-agnostic — a Python backend, a Java API, or a Node.js server all speak JSON. It is the universal language of data exchange on the internet, and every web API you will ever call will respond in JSON.

JavaScript Object

Keys have no quotes (or single quotes). Can hold functions, dates, undefined. Only lives in memory.
Example: `{ title: 'Instagram', clicks: 234, isActive: true }`

JSON String

All keys **MUST** be double-quoted. No functions, no undefined. Plain text — can be stored in files, databases, or sent over a network. Example: `{"title": "Instagram", "clicks": 234, "isActive": true}`

02 JSON.stringify() & JSON.parse()

These two methods are your translation tools between JavaScript objects (which live in memory) and JSON strings (which can be stored or transmitted). `JSON.stringify()` converts a JS value into a JSON string — use this before saving to `localStorage` or sending to an API. `JSON.parse()` converts a JSON string back into a live JavaScript object — use this after reading from `localStorage` or receiving an API response. Forgetting to parse or stringify is one of the most common beginner bugs.

```
const links = [
  { title: 'Instagram', url: 'https://instagram.com/aarya', clicks: 234 },
  { title: 'YouTube', url: 'https://youtube.com/@aarya', clicks: 512 }
];
const jsonString = JSON.stringify(links);
console.log(typeof jsonString);
```

JavaScript array of objects (lives in memory)

Converts array to JSON text — now safe to store 'string' — it is now plain text, not an object

```
const parsedLinks = JSON.parse(jsonString);
console.log(parsedLinks[0].title);
```

Converts JSON text back to a live JS array
'Instagram' — fully usable object again

03 localStorage — Browser-Side Storage

CONCEPT — What is localStorage?

localStorage is a simple key-value store built into every modern browser. Data saved there persists even after the tab is closed, the browser is restarted, or the computer is turned off — until the user manually clears their browser data or you programmatically delete it. It stores only strings, so you must JSON.stringify() objects before saving and JSON.parse() them after reading. Think of localStorage as a small personal notebook the browser keeps for each website — separate for each domain, inaccessible to other websites.

```
// SAVE — store the links array
localStorage.setItem('bioLinks', JSON.stringify(links));
    Key: 'bioLinks', Value: JSON string of the array
// READ — retrieve the links array
const savedData = localStorage.getItem('bioLinks');
    Returns the JSON string (or null if key doesn't exist)
const savedLinks = savedData ? JSON.parse(savedData) : [];
    Parse if data exists; otherwise default to empty array
// UPDATE — read, modify, save back
savedLinks.push({ title: 'GitHub', url: 'https://github.com/aarya', clicks: 0 });
    Add new link to the parsed array
localStorage.setItem('bioLinks', JSON.stringify(savedLinks));
    Save the updated array back
// DELETE — remove a key entirely
localStorage.removeItem('bioLinks');
    Deletes the 'bioLinks' key from storage
localStorage.clear();
    Deletes ALL keys stored by this site — use carefully
```

TRY THIS LIVE

Wire localStorage into your Bio-Link App: (1) On page load (DOMContentLoaded event), read from localStorage and render existing links. (2) Every time a link is added, save the updated links array back to localStorage. (3) Every time a link is deleted, save the updated array again. Test: Add 3 links -> close the tab -> reopen -> your links should still be there. This is persistence — your app now remembers data.

04 Practical Patterns & Safety

```
function saveLinks(linksArray) {
  localStorage.setItem('bioLinks', JSON.stringify(linksArray));
}
function loadLinks() {
  try {
    const data = localStorage.getItem('bioLinks');
    return data ? JSON.parse(data) : [];
  } catch (e) {
    console.error('Could not load links:', e);
    return [];
  }
}
```

Centralise save logic in one function

Centralise load logic — safer with try/catch

Return empty array if nothing saved yet

JSON.parse fails if stored string is corrupted

Fail gracefully — never crash the app

PRO TIP — localStorage Limitations

localStorage has a 5MB limit per origin. It is synchronous — reading or writing large amounts of data can block the main thread and freeze the UI. It is not accessible from Web Workers. Most importantly, it is not a database — it cannot be queried, indexed, or accessed from a server. Use it only for small, non-sensitive UI preferences (theme, language) or prototype persistence. For real apps, data lives in a backend database (MySQL, Firestore) — which you will use from React Advanced onwards.

WARNING — Never Store Sensitive Data in localStorage

localStorage is accessible to any JavaScript running on your page — including third-party scripts and, if your site has an XSS vulnerability, to attackers. Never store passwords, OTPs, payment details, or authentication tokens in localStorage. Authentication tokens should go in httpOnly cookies — a topic you will cover when you build authentication in Module 3.

WHERE TO GO NEXT

Your Bio-Link App now persists data across reloads. A user can close the browser and come back to find their links exactly as they left them — which is the minimum bar for a usable web app.

Explore deeper — MDN Docs: 'Web Storage API' —
developer.mozilla.org/en-US/docs/Web/API/Web_Storage_API

Try on your own — Add a dark mode toggle to your app. Save the user's preference ('dark' or 'light') in localStorage under the key 'theme'. On page load, read the saved theme and apply it. The preference should survive page reloads.

Connect the dots — Session 10 introduces ES6 features including template literals, destructuring, and spread — modern syntax that will make your localStorage read/write code significantly shorter and more

readable.

Going further — Research IndexedDB — the browser's full transactional database for larger structured data. Libraries like Dexie.js wrap it in a friendly API. Also look at sessionStorage, which is identical to localStorage but is cleared when the tab closes.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript tutor. Explain what localStorage is and why it is useful for web apps in simple Hinglish. Use the analogy of a shopkeeper in Surat who writes down a customer's preferences in a small diary so he remembers them next time. Keep it under 150 words.

SESSION

10

60 MIN · MODERN JAVASCRIPT

ES6 Syntax & Modern Features

◆ Learning Objectives

- Use template literals to build dynamic strings without concatenation
- Apply destructuring to extract values from arrays and objects cleanly
- Use the spread operator to copy and merge arrays and objects
- Use the rest operator to collect variable function arguments
- Set default parameter values in function definitions
- Refactor Bio-Link App code to use modern ES6 syntax throughout

01 Why ES6 Matters

ES6 (ECMAScript 2015) was the biggest update to JavaScript since its creation. It introduced cleaner, shorter, and more expressive syntax that is now the industry standard. Every job description for a React developer lists ES6 as a requirement. All the features in this session — template literals, destructuring, spread, rest — are not optional extras; they are the daily vocabulary of modern JavaScript development. React code uses these so heavily that trying to read React without knowing ES6 is like trying to read a Gujarati newspaper without knowing the script.

02 Template Literals

CONCEPT — Template Literals

Template literals use backticks (`) instead of quotes and allow embedding expressions directly inside strings using `\${expression}`. They also support multi-line strings without escape characters. Before ES6, building strings from variables required messy concatenation with + signs. With template literals, you write the string naturally and drop variables inline — identical to how Python f-strings and PHP's heredoc strings work.

```
const userName = 'Aarya';
const followers = 1500;
// Old way – error-prone concatenation
console.log('Hello ' + userName + ', you have ' + followers + ' followers!');
// ES6 template literal – clean and readable
console.log(`Hello ${userName}, you have ${followers} followers!`);

  Embed variables with ${}
console.log(`Engagement: ${((234/followers)*100).toFixed(1)}%`);

  Embed full expressions — not just variables
const card = `
  <div class='link-card'>
```

Multi-line string — no \n escape needed

```
<a href='${url}'>${title}</a>
</div>
`;
```

This HTML string can be used with innerHTML

03 Destructuring — Unpacking Values Cleanly

Destructuring is a shorthand syntax that extracts values from arrays or objects into named variables in a single line. Without destructuring, you need multiple lines to pull properties out of an object. With it, you unpack everything in one clean statement. This is used constantly in React: every time you write `const [count, setCount] = useState(0)` or `const { title, url } = props` — that is destructuring.

```
// Object destructuring
const link = { title: 'Instagram', url: 'https://instagram.com/aarya', clicks: 234 };
const { title, url, clicks } = link;           Extract three properties into three variables in ONE line
console.log(title, url, clicks);               'Instagram', 'https://...', 234
const { title: linkName, clicks: totalClicks = 0 } = link;
    Rename property (title -> linkName) and set default (0 if clicks missing)
// Array destructuring
const colors = ['#E1306C', '#FF0000', '#0A66C2'];
const [instagramColor, youtubeColor, linkedinColor] = colors;
    Position-based extraction
const [first, , third] = colors;               Skip middle element with empty comma
```

04 Spread Operator — Expanding Collections

CONCEPT — Spread Operator (...)

The spread operator (...) expands an array or object into its individual elements. Use it to copy arrays, merge arrays, copy objects, or merge objects — all without mutating the originals. This is the React-safe way to add items to arrays and update object properties. Instead of `links.push(newLink)` which mutates, you write `[...links, newLink]` which creates a new array — this is critical for React's state management system to detect changes.

```
const links = [{title:'Instagram'}, {title:'YouTube'}];
const newLink = {title:'LinkedIn', url:'...', clicks:0};
// Adding to array (React-safe — creates NEW array)
const updatedLinks = [...links, newLink];           Spreads existing links + appends newLink
// Merging two arrays
const featuredLinks = [{title:'GitHub'}];
```

```
const allLinks = [...links, ...featuredLinks];
```

Combines both arrays into one new array

```
// Copying an object
```

```
const linkCopy = { ...newLink };
```

Shallow copy — safe to modify without affecting original

```
// Updating one property (React pattern)
```

```
const updatedLink = { ...newLink, clicks: newLink.clicks + 1 };
```

New object with same properties but incremented clicks

05 Rest Parameters & Default Parameters

```
// Rest parameters — collects remaining args into an array
```

```
const createProfile = (name, bio, ...socialLinks) => {
```

...socialLinks collects all arguments from the third onward

```
  return { name, bio, links: socialLinks };    Shorthand property: name = name: name
```

```
};
```

```
createProfile('Aarya', 'Creator', 'instagram.com', 'youtube.com', 'linkedin.com');
```

socialLinks = ['instagram.com', 'youtube.com', 'linkedin.com']

```
// Default parameters
```

```
const formatLink = (title, color = '#333', clicks = 0) => {
```

color and clicks have defaults

```
  return `${title} | ${color} | ${clicks} clicks`;
```

```
};
```

```
formatLink('Instagram', '#E1306C');
```

clicks defaults to 0 — 'Instagram | #E1306C | 0 clicks'

TRY THIS LIVE

Refactor your Bio-Link App's addLink and removeLink functions from Session 08 to use ES6 throughout: use destructuring when reading link properties, template literals in console messages, spread when adding to the links array (no more push — create a new array), and shorthand property names when creating link objects. The behaviour should be identical but the code should be noticeably cleaner.

PRO TIP — Shorthand Property Names

When a variable name matches the object property name you want, ES6 lets you write just the variable name once. Instead of { title: title, url: url, clicks: clicks } you write { title, url, clicks }. This shorthand is used in virtually every modern JavaScript codebase and every React component. Get comfortable with it now.

WHERE TO GO NEXT

You now write modern, professional-grade JavaScript. Every ES6 feature in this session will appear in React code daily — often in the same line.

Explore deeper — Explore.js (JavaScript features explorer): exploringjs.com/es6/ by Dr. Axel Rauschmayer — the definitive reference for ES6+ features with clear examples

Try on your own — Rewrite your `loadLinks()` and `saveLinks()` functions from Session 09 using destructuring and template literals. Aim to reduce the total line count without losing any functionality.

Connect the dots — Session 11 introduces async JavaScript — Promises and `async/await` — which are also ES6/ES2017 features. The same modern-syntax mindset you built today applies directly to asynchronous patterns.

Going further — Look up 'Optional chaining (`?.`) and nullish coalescing (`??`)' — ES2020 features. Also explore 'logical assignment operators' (`||=`, `&&=`, `??=`) from ES2021, now widely used in production React codebases.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript tutor. Explain the ES6 spread operator in simple Hinglish. Use the analogy of mixing ingredients from two different dabba tiffins into one new container. Show how spread can merge two arrays and two objects. Keep it under 150 words.

SESSION

11

60 MIN · ASYNC JAVASCRIPT

Asynchronous JS – Callbacks, Promises & Async/Await

◆ Learning Objectives

- Explain why JavaScript is single-threaded and what that means for UI responsiveness
- Distinguish between synchronous (blocking) and asynchronous (non-blocking) code
- Use `setTimeout` and `setInterval` to schedule deferred code execution
- Explain callback functions and the callback hell problem
- Create, resolve, and reject Promises and chain them with `.then()` and `.catch()`
- Write async functions using `async/await` syntax for cleaner asynchronous code

01 The Single Thread Problem

JavaScript runs on a single thread — it can only do one thing at a time. This means it cannot make a network request and simultaneously respond to a button click. If JS had to wait (block) for every slow operation — loading an image, fetching data from an API, reading a file — the entire page would freeze until each task finished. On a slow Indian mobile network, a blocked fetch could freeze your app for 3-5 seconds. Asynchronous JavaScript solves this: slow tasks are handed off and JS moves on, returning to handle the result when it is ready — without ever blocking the UI.

CONCEPT — The Event Loop

JavaScript's event loop is what makes async possible. When an async operation (network request, timer, file read) is started, it is handed to the browser's Web APIs, which handle it in the background. Meanwhile, the JS engine continues executing the rest of the script. When the async operation finishes, its callback is placed in the task queue. The event loop checks: 'Is the call stack empty?' If yes, it picks the next task from the queue and executes it. This cooperative scheduling model is how Node.js handles thousands of simultaneous connections on a single thread.

02 setTimeout & setInterval — Scheduled Execution

```
console.log('1. App starting...');
setTimeout(() => {
  console.log('3. Loading complete!');
}, 2000);
console.log('2. Showing loading spinner...');
// Output order: 1, 2, 3 – not 1, 3, 2
const timer = setInterval(() => {
  console.log('Checking for new notifications...');
```

Runs immediately — synchronous

Async — hands off to browser, does NOT block

Runs AFTER the current script finishes — even with 0ms delay

2000ms = 2 seconds delay

Runs BEFORE the timeout callback — JS did not wait

Repeats every 1 second indefinitely

```
}, 1000);
clearInterval(timer);
```

Stop the repeating interval — always clear when done

03 Callbacks & Callback Hell

The original async pattern in JavaScript was callbacks — passing a function as an argument to be called when an async operation completes. Simple callbacks work well for one level of async. But when you need to chain multiple async operations — load the user profile, then fetch their links, then load the clicks for each link — callbacks nest inside callbacks inside callbacks. This deeply nested structure is called callback hell, and it makes code impossible to read, debug, or maintain.

```
// Callback hell — three nested async operations
```

```
loadUser(userId, function(user) {
  loadLinks(user.id, function(links) {
    loadClicks(links[0].id, function(clicks) {
      console.log('Clicks:', clicks);
    });
  });
});
```

Step 1: load user

Step 2: load their links (nested inside step 1's callback)

Step 3: load clicks (nested inside step 2's callback)

Finally — 3 levels deep

The 'pyramid of doom' — gets worse with error handling

04 Promises — A Better Async Pattern

CONCEPT — What is a Promise?

A Promise is an object that represents the eventual result of an async operation. It starts in a 'pending' state and moves to either 'fulfilled' (success) or 'rejected' (failure) — exactly once, never both. You attach handlers using `.then()` for success and `.catch()` for errors. Promises can be chained — `.then()` returns a new Promise, so you can sequence async operations in a flat, readable chain instead of nesting callbacks. Think of a Promise like ordering food on Swiggy: you get an order confirmation immediately (pending) — later the state changes to delivered (fulfilled) or cancelled (rejected).

```
const loadUserProfile = (userId) => {
  return new Promise((resolve, reject) => {

    setTimeout(() => {
      if (userId > 0) {
```

Constructor takes two callbacks: resolve (success) and reject (failure)

Simulate a 1-second async operation

```

resolve({ name: 'Aarya', followers: 1500 });
  // Fulfills the Promise with user data
} else {
  reject(new Error('Invalid user ID')); // Rejects with an error
}
}, 1000);
});
};

loadUserProfile(1)
  .then(user => console.log(`Welcome, ${user.name}!`))
  // .then handles success
  .catch(err => console.error('Error:', err.message));
  // .catch handles rejection

```

05 async / await — Synchronous-Looking Async Code

async/await is syntax sugar built on top of Promises. The async keyword marks a function as asynchronous — it always returns a Promise. Inside an async function, the await keyword pauses execution until the awaited Promise settles, then returns its resolved value. The code reads like synchronous top-to-bottom logic, dramatically improving readability. This is now the standard way to write async code in modern JavaScript and is what you will use in every React component that fetches data.

```

const fetchData = async (userId) => {
  try {
    const user = await loadUserProfile(userId);
    // await pauses here until Promise resolves — no .then() needed
    console.log(`Welcome, ${user.name}!`);
    const links = await loadLinks(user.id);
    // Reads like synchronous code — clean and linear
    // Await the next async step — perfectly sequential
    console.log('Links loaded:', links.length);
  } catch (error) {
    // Catches any rejection in any of the awaited calls
    console.error('Something failed:', error.message);
  }
};

fetchData(1);
// Call the async function — returns a Promise immediately

```

TRY THIS LIVE

In your browser console, write an async function called simulateLoading that: (1) logs 'Fetching your bio links...', (2) awaits a 2-second delay (create a helper function delay(ms) that returns new Promise(resolve => setTimeout(resolve, ms))), (3) logs 'Links loaded! Displaying...', (4) logs each link from your links array. Call simulateLoading() and watch the timing in the console. This is the exact pattern real API calls follow.

PRO TIP — Promise.all for Parallel Async

If you have multiple independent async calls, do not await them sequentially — each would wait for the previous to finish. Use `Promise.all([p1, p2, p3])` to run them all in parallel and wait for all to complete. This can reduce total wait time dramatically: three 1-second calls sequentially take 3 seconds; in parallel they take ~1 second. This matters significantly for users on slow Indian mobile networks.

WHERE TO GO NEXT

You now understand the async model that powers every API call, every real-time update, and every loading spinner in modern web apps. This is one of the hardest concepts in JavaScript — revisit this session if anything feels unclear.

Explore deeper — JavaScript.info: 'Promises, async/await' — javascript.info/async — the clearest written explanation of the async model with interactive examples

Try on your own — Chain three Promises without `async/await`, then rewrite the same chain using `async/await`. Compare the two versions side by side and notice the difference in readability.

Connect the dots — Session 12 puts `async/await` into action with the real `fetch()` API — you will call a live public API and display real data in your app for the first time.

Going further — Look up 'Promise combinators': `Promise.all` (all must succeed), `Promise.allSettled` (all run regardless of failure), `Promise.race` (first to finish wins), and `Promise.any` (first to succeed wins). Each is useful in different multi-fetch scenarios.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript teacher. Explain Promises in simple Hinglish using the analogy of ordering a Swiggy food delivery: pending (order placed), fulfilled (food arrived), rejected (order cancelled). Then show the `.then()` and `.catch()` pattern in 5 lines of code. Keep it under 150 words.

SESSION
12

60 MIN · ASYNC JAVASCRIPT

Asynchronous JS (Part 2) – Fetch API

◆ Learning Objectives

- Explain what an HTTP request is and how the request-response cycle works
- Use the Fetch API to make GET requests and handle the response
- Parse JSON responses from an API using `response.json()`
- Make POST requests using `fetch` with a body and headers
- Handle loading states and API errors gracefully in the UI
- Fetch real mock user data and display it in the Bio-Link App

01 How the Web Communicates — HTTP Basics

Every piece of data your app shows to users — a product listing, a weather report, a social feed — was fetched from a server using HTTP (HyperText Transfer Protocol). Your browser sends an HTTP request with a method (GET, POST, PUT, DELETE), a URL, optional headers, and optional body. The server responds with a status code (200 = OK, 404 = Not Found, 500 = Server Error) and a response body, almost always in JSON format. Understanding this request-response cycle is essential — it is the same mechanism used by Swiggy fetching restaurant menus, IRCTC loading train schedules, and Google Maps loading route data.

HTTP Method	Purpose	Example Use in Bio-Link App
GET	Retrieve data — no body	Load all saved links for a user
POST	Create new data — sends body	Add a new social link to the server
PUT	Replace entire resource	Replace all properties of an existing link
PATCH	Update partial resource	Update only the click count of one link
DELETE	Remove resource	Delete a specific social link

02 The Fetch API — GET Request

The Fetch API is built into modern browsers — no library installation needed. It returns a Promise that resolves to a Response object. Critically, you must call `response.json()` to parse the body — and this is itself an async operation returning another Promise, which is why you need two awaits in sequence. The demo URL

<https://jsonplaceholder.typicode.com/users> is a free fake REST API used by thousands of developers worldwide for testing — perfect for this session.

```
const fetchUsers = async () => {
  try {
    const response = await fetch('https://jsonplaceholder.typicode.com/users');
    if (!response.ok) {
      throw new Error(`HTTP error: ${response.status}`);
    }
    const users = await response.json();
    console.log('Users loaded:', users.length);
    users.forEach(user => console.log(`${user.name} - ${user.email}`));
  } catch (error) {
    console.error('Fetch failed:', error.message);
  }
};
fetchUsers();
```

Always wrap fetch in try/catch

First await: sends GET request and waits for initial response

Check HTTP status — fetch only rejects on network failure, not 404/500

Manually throw for bad status codes

Second await: parses JSON body from the response stream

Returns 10 users from the fake API

Loop and display

Call — open Network tab in DevTools to watch the request

TRY THIS LIVE

Open browser DevTools (F12) and go to the Network tab. Paste the `fetchUsers` function into the Console and call it. Watch the GET request appear in the Network tab — click on it to see the full request URL, headers, status code (200 OK), and the JSON response body. This is exactly what happens every time a React app fetches data — you are now seeing the underlying mechanism.

03 POST Request — Sending Data to a Server

```
const createLink = async (linkData) => {
  try {
    const response = await fetch('https://jsonplaceholder.typicode.com/posts', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(linkData)
    });
    const result = await response.json();
  }
};
```

Second argument: request options object

Override default GET method

Tell server we are sending JSON

Serialise object to JSON string — server receives this

Parse server's response (usually the created object with an id)

```

    console.log('Created:', result);
  } catch (error) {
    console.error('POST failed:', error.message);
  }
};
createLink({ title: 'Instagram', url: 'https://instagram.com/aarya', clicks: 0 });
// jsonplaceholder simulates a 201 Created response

```

04 Loading States & UI Feedback

Real apps always show visual feedback during async operations — a spinner, a skeleton screen, a 'Loading...' text. Without this, users click buttons multiple times (causing duplicate submissions) or think the app has crashed. Managing loading state is simple: set a flag to true before the fetch, false after it completes (whether success or error), and update the UI accordingly. This exact pattern — `isLoading` state — is one of the first things you will implement in React using `useState`.

```

let isLoading = false; // Loading state variable
const loadingEl = document.querySelector('#loading-msg');
// A <p id='loading-msg'> in your HTML
const fetchAndDisplay = async () => {
  isLoading = true;
  loadingEl.textContent = 'Fetching your links...';
  // Show loading message
  loadingEl.style.display = 'block';
  try {
    const response = await fetch('https://jsonplaceholder.typicode.com/users/1');
    const user = await response.json();
    document.querySelector('#profile-name').textContent = user.name;
    // Update UI on success
  } catch (e) {
    loadingEl.textContent = 'Failed to load. Please try again.';
    // User-friendly error message
  } finally {
    // Runs whether success or error
    isLoading = false;
    loadingEl.style.display = 'none'; // Always hide the loading message at the end
  }
};

```

PRO TIP — Disable Buttons During Fetch

Always disable the submit button while a fetch is in progress using `button.disabled = true`. Enable it again in the finally block. Without this, impatient users click the button repeatedly, triggering multiple simultaneous API

calls — a problem called a 'double submit' that can create duplicate records in your database. This is a real production bug that has caused data issues on Indian fintech apps processing payments.

CASE STUDY — PhonePe (Bengaluru / Ahmedabad users)

Context — PhonePe is India's largest UPI payments platform with over 500 million registered users, widely used across Gujarat for daily transactions.

Challenge — The UPI money transfer flow requires fetching receiver account details (name, bank) before confirming a transfer. Any loading delay without feedback caused users to tap 'Pay' multiple times — risking duplicate payments.

What they did — PhonePe's frontend team implemented strict loading state management: the Pay button disables immediately on tap, a spinner replaces the button text, and the button re-enables only after the transaction completes or fails — with a clear status message either way.

Result — Duplicate payment incidents dropped significantly, and user trust in the transaction flow improved — shown by lower drop-off rates at the confirmation step.

Takeaway — Loading state and button disabling during fetch is not polish — it is correctness. Implement it from day one.

Source: PhonePe Engineering Blog, 2022; UPI transaction flow analysis

WHERE TO GO NEXT

You can now retrieve real data from the internet and display it in your app. This bridges the gap from a local prototype to a connected web application.

Explore deeper — JSONPlaceholder docs: jsonplaceholder.typicode.com — explore all available endpoints (users, posts, comments, albums) for practice. Also see Reqres.in for more realistic fake auth APIs.

Try on your own — Fetch the user with ID 1 from <https://jsonplaceholder.typicode.com/users/1> and display their name, email, and username in your Bio-Link App's profile section. Show a loading state while fetching.

Connect the dots — Session 13 covers error handling in depth — you will learn to throw, catch, and display meaningful errors for all the async operations you now know how to make.

Going further — Research Axios — the most popular HTTP client library for JavaScript. It automatically parses JSON, has built-in timeout support, and provides cleaner error objects than raw fetch. You will use it in React Fundamentals (Module 2, Session 11).

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript instructor. Explain the Fetch API in simple Hinglish. Use the analogy of ordering from a restaurant through a delivery app: the GET request is like checking the menu, the POST request is like placing an order. Show 3 lines of fetch code. Keep it under 150 words.

SESSION

13

60 MIN · DEFENSIVE CODING

Error Handling in JS

◆ Learning Objectives

- Distinguish between syntax errors, runtime errors, and logical errors
- Use try / catch / finally blocks to handle runtime exceptions
- Throw custom errors using the throw keyword and the Error class
- Create custom Error subclasses for domain-specific error types
- Handle async errors in Promises using .catch() and in async/await using try/catch
- Validate user inputs and throw meaningful errors before any API call is made

01 Types of Errors in JavaScript

JavaScript programs fail in three distinct ways. Syntax errors are caught before the code runs — a missing bracket, a typo in a keyword. Runtime errors occur during execution — calling a method on undefined, dividing by zero, trying to access a property of null. Logical errors are the hardest: the code runs without crashing, but produces the wrong result. A logical error in a billing calculation could silently overcharge lakhs of users before anyone notices. Robust error handling primarily targets runtime errors — but rigorous testing catches logical ones.

Error Type	When It Occurs	Example
SyntaxError	Before code runs (parsing)	let x = ; (missing value)
ReferenceError	Accessing an undeclared variable	console.log(userName) before let userName
TypeError	Wrong type for an operation	null.toUpperCase() — cannot call method on null
RangeError	Value out of valid range	new Array(-1) — negative array length
URIError	Malformed URI string	decodeURIComponent('%')
Custom Error	Application logic violation	throw new ValidationError('URL is required')

02 try / catch / finally

CONCEPT — try / catch / finally

Wrapping code in a try block lets you intercept runtime errors instead of crashing the entire program. If any

line inside the try block throws an error, JS immediately jumps to the catch block — skipping remaining try code — and passes the error object as the catch parameter. The finally block always runs after try and catch, regardless of whether an error occurred. Use finally for cleanup code that must always execute: hiding a loading spinner, re-enabling a button, closing a file.

```
const getUsername = (user) => {
  try {
    const upper = user.name.toUpperCase();

    return upper;
  } catch (error) {
    console.error('Error reading user name:', error.message);
    error.message contains the human-readable description
    return 'Anonymous';

  } finally {
    console.log('getUsername call finished.');
```

Attempt this code — if any line throws, jump to catch

Throws TypeError if user is null or has no name property

error is the thrown Error object

Graceful fallback — function still returns something useful

Runs no matter what

Good for audit logging or cleanup

```
  }
};
getUsername(null);
```

Triggers TypeError — caught gracefully

03 throw — Raising Your Own Errors

The throw keyword lets you manually signal an error from inside your code. You can throw any value — a string, a number, or most usefully, an Error object. Throwing Error objects is strongly preferred because they include a message and a stack trace — a list of function calls that led to the error — which makes debugging vastly easier. A function should throw when it receives invalid input it cannot safely handle, rather than silently returning wrong results.

```
const validateURL = (url) => {
  if (!url || url.trim() === '') {
    throw new Error('URL is required — please enter a social media link.');
```

Validation function — throws if invalid

Empty or missing URL

Throws with clear user-facing message

```
  }
  if (!url.startsWith('https://')) {
    throw new Error('URL must start with https:// for security.');
```

Security check — enforce HTTPS

```
  }
  return url.trim();
};
try {
```

Return sanitised URL if all checks pass

```
const cleanUrl = validateURL('http://instagram.com/aarya');
// Throws because URL starts with http not https
} catch (e) {
document.querySelector('#error-msg').textContent = e.message;
// Show the thrown message in the UI
}
```

04 Custom Error Classes

```
class ValidationError extends Error {
  constructor(message, field) {
    super(message);
    this.name = 'ValidationError';
    this.field = field;
  }
}

try {
  throw new ValidationError('URL cannot be empty', 'url');
  // Throw with field information
} catch (e) {
  if (e instanceof ValidationError) {
    console.log(`Field '${e.field}' failed: ${e.message}`);
    // Targeted response for validation errors
  } else {
    console.error('Unexpected error:', e);
    // Rethrow or log unexpected errors differently
  }
}
```

TRY THIS LIVE

Update your Bio-Link App's addLink function to validate inputs before adding. Throw a ValidationError if the title is empty, if the URL is empty, or if the URL does not start with 'https://'. Wrap the function call in a try/catch that displays the error.message in a red error paragraph below the form. Test all three invalid cases. This is production-quality input validation.

05 Error Handling in Async Code

CONCEPT — Async Error Handling

Unhandled Promise rejections are silent failures that crash Node.js servers and produce confusing console warnings in browsers. In async/await code, always use try/catch to handle rejections. In Promise chain code, always end with .catch(). Never leave a Promise rejection unhandled. Also note: fetch() does NOT reject on HTTP error codes (404, 500) — it only rejects on network failure. You must manually check response.ok and throw if false, as shown in Session 12.

```
// Promise chain — .catch() at the end
fetch('https://api.example.com/links')
.then(res => { if (!res.ok) throw new Error(res.status); return res.json(); })
  Manual HTTP error check inside .then()
  .then(data => console.log(data))
.catch(err => console.error('Request failed:', err.message));
  Catches both network failures AND thrown HTTP errors

// async/await — try/catch
const loadData = async () => {
  try {
    const res = await fetch('https://api.example.com/links');
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const data = await res.json();
    return data;
  } catch (err) {
    console.error('Failed to load:', err.message);
  }
};
```

PRO TIP — Error Messages for Users vs Developers

Always maintain two types of error messages: a technical message for developers (logged to console with full stack trace) and a user-friendly message for the UI (clear, actionable, no jargon). Never display a raw JS error in the UI — 'TypeError: Cannot read properties of null' is meaningless and frightening to a user. Instead show: 'We could not load your links. Please check your internet connection and try again.'

WHERE TO GO NEXT

Defensive error handling separates beginner code from production-ready code. Every function you write from here should anticipate failure and handle it gracefully.

Explore deeper — MDN Docs: 'Error' —

developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Error — read about all built-in error types and the stack property

Try on your own — Add a global error handler using `window.addEventListener('error', handler)` and `window.addEventListener('unhandledrejection', handler)` to catch any errors your try/catch blocks miss. Log

them to the console with a clear label.

Connect the dots — Session 14 covers modular JavaScript — splitting your now-complex app code into separate files so each concern (validation, DOM, storage) lives in its own clean module.

Going further — Research 'Sentry' — the industry-standard error monitoring service used by companies like Flipkart and Razorpay. It captures uncaught JS errors in production and sends them to a dashboard. Free tier is available for personal projects.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript debugging coach. Explain the difference between a try-catch block and an if-else block in Hinglish. Use the analogy of a quality inspector at a Surat textile mill who catches defective fabric (catch) versus a guard at the gate who checks credentials before entry (if-else). Keep it under 150 words.

SESSION

14

60 MIN · CODE ARCHITECTURE

Modular JavaScript – Imports & Exports◆ **Learning Objectives**

- Explain why modular code is better than a single large file
- Export functions and variables using named exports and default exports
- Import named and default exports into other JavaScript files
- Understand the difference between ES Modules and CommonJS (require/module.exports)
- Restructure the Bio-Link App into separate modules by responsibility
- Understand how ES Modules directly prepare you for React's component architecture

01 Why Modules Exist — The Problem with One Big File

When your Bio-Link App was new, all code in one file made sense. Now it has validation logic, DOM manipulation, localStorage operations, fetch calls, and error handling — all mixed together. Finding the right function means scrolling through hundreds of lines. Changing one function risks breaking something unrelated. This is why every professional JavaScript project is split into modules — separate files where each file has a single, clear responsibility. React takes this further: every component is its own module file. Learning modules now makes the React file structure immediately familiar.

CONCEPT — ES Modules vs CommonJS

There are two module systems in JavaScript. CommonJS (require/module.exports) was invented for Node.js and runs only in server environments. ES Modules (import/export) are the official browser standard from ES6 and work in both browsers and Node.js (v12+). In modern front-end development and in React, you always use ES Modules. When you see require() in code, that is a Node.js or older codebase. When you see import, that is modern ES Module syntax.

02 Named Exports & Named Imports

Named exports let you export multiple values from a single file — each with its own name. The consumer imports them using the exact same name in curly braces. A file can have as many named exports as needed. This is ideal for utility files that contain many related helper functions, like a validation module or a formatting module.

```
// utils.js – utility/helper module
export const validateURL = (url) => {           Named export — export keyword before const
  if (!url || !url.startsWith('https://')) throw new Error('Invalid URL');
  return url.trim();
};
```

```
export const formatLink = (title, url) => `■ ${title} -> ${url}`;
```

Another named export in the same file

```
export const generateId = () => Date.now().toString(36);
```

Third named export — generates a short unique ID

```
// main.js — consuming the module
```

```
import { validateURL, formatLink } from './utils.js';
```

Named imports — must match export names exactly

```
import { validateURL as validate } from './utils.js';
```

Rename on import with 'as' alias

03 Default Exports & Default Imports

CONCEPT — Default Export

A file can have at most ONE default export — typically the main thing the file is about. When importing a default export, you choose the name yourself (no curly braces). React component files almost always use default exports: `export default function App() {}` — and you import it as `import App from './App'`. Utility files with many helpers use named exports. Knowing when to use which is a hallmark of well-organised JS codebases.

```
// storage.js — localStorage module
```

```
const STORAGE_KEY = 'bioLinks';
```

Private to this module — not exported

```
export const saveLinks = (links) =>
```

Named export

```
localStorage.setItem(STORAGE_KEY, JSON.stringify(links));
```

```
export const loadLinks = () => {
```

Named export

```
const data = localStorage.getItem(STORAGE_KEY);
```

```
  return data ? JSON.parse(data) : [];
```

```
};
```

```
export default { saveLinks, loadLinks };
```

Default export — the whole module as one object

```
// main.js
```

```
import storage from './storage.js';
```

Import default — name is your choice

```
import { saveLinks, loadLinks } from './storage.js';
```

OR import named exports individually

04 Restructuring the Bio-Link App

A well-structured module layout for the Bio-Link App separates concerns cleanly. Each file has one job: `utils.js` handles validation and formatting, `storage.js` handles `localStorage`, `api.js` handles fetch calls, `dom.js` handles rendering and DOM manipulation, and `main.js` is the entry point that wires everything together. This is almost identical

to how a React project organises its files — components, hooks, services, and utils are all separate folders with distinct responsibilities.

File	Responsibility	Exports
utils.js	Validation, string formatting, ID generation	validateURL, formatLink, generateId
storage.js	localStorage read / write / clear	saveLinks, loadLinks, clearLinks
api.js	All fetch() calls to external APIs	fetchUser, postLink, deleteLink
dom.js	createElement, render, update DOM	renderLinkCard, showError, setLoading
main.js	Entry point — wires modules together on DOMContentLoaded	No exports — side-effects only

TRY THIS LIVE

Restructure your Bio-Link App: create utils.js, storage.js, and dom.js files. Move your validation functions to utils.js, localStorage functions to storage.js, and createLinkCard / renderLinks to dom.js. Export each function. In main.js, import and use them. Add type='module' to your script tag in index.html: `<script type='module' src='main.js'></script>`. This attribute enables ES Modules in the browser. Verify the app still works exactly as before.

PRO TIP — Module Scope is Private by Default

Unlike regular script tags which share a global scope (variables leak between files), ES Modules have their own private scope. A variable declared in utils.js is not accessible in storage.js unless explicitly exported. This private-by-default design prevents naming conflicts, accidental overwrites, and the type of global variable bugs that plagued large jQuery-era codebases. It is one of the biggest quality-of-life improvements in modern JavaScript.

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Programming in Modern JavaScript — Lecture on Modules and Code Organisation (IIT Bombay NPTEL course)

Link — <https://nptel.ac.in/courses/106101238>

Always link the official channel — beware fan-made or re-upload copies.

WHERE TO GO NEXT

Your Bio-Link App is now professionally structured — each concern in its own file, each file imported where needed. This is exactly how real-world projects are organised.

Explore deeper — MDN Docs: 'JavaScript modules' —

developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Modules — covers import maps, dynamic imports, and module bundlers

Try on your own — Add a fourth module: `config.js` that exports constants like `APP_NAME`, `VERSION`, and `STORAGE_KEY`. Import these in `main.js` and `storage.js`. Notice how centralising constants prevents typos when the same string is used in multiple files.

Connect the dots — Session 15 introduces Classes and OOP — you will add a `LinkManager` class that encapsulates all link operations (add, remove, save, load) using the module structure you just built.

Going further — Research 'dynamic imports' — `import('./module.js').then()` — which loads a module only when needed, reducing initial page load time. This is how React's `lazy()` and `Suspense` work under the hood.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a software architecture tutor. Explain JavaScript modules and the export/import system in simple Hinglish. Use the analogy of different departments in a company like TCS — HR, Finance, Engineering each handle their own work and share only what is needed. Keep it under 150 words.

SESSION

15

60 MIN · OBJECT-ORIENTED PROGRAMMING

Classes & OOP in JS

◆ Learning Objectives

- Explain the four pillars of OOP: encapsulation, abstraction, inheritance, and polymorphism
- Declare a class with a constructor, instance properties, and instance methods
- Use the extends keyword to create subclasses that inherit from parent classes
- Call parent class constructors and methods using super
- Use static methods for utility functions that belong to the class, not instances
- Build a UserProfile class for the Bio-Link App that encapsulates all profile behaviour

01 OOP — Organising Code Around Real-World Entities

Object-Oriented Programming (OOP) is a programming paradigm where code is organised around objects — entities that combine related data (properties) and behaviour (methods). Instead of having separate variables for a user's name, followers, and links, plus separate functions to add and remove links, you bundle all of it inside a UserProfile class. This mirrors how we think about the real world: a user IS a thing with attributes and capabilities, not a scattered collection of loose variables and functions.

CONCEPT — The Four Pillars of OOP

Encapsulation: bundling data and methods together, hiding internal implementation. Abstraction: exposing only what the outside world needs, hiding complexity. Inheritance: a child class inherits properties and methods from a parent class, avoiding duplication. Polymorphism: subclasses can override parent methods to provide different behaviour while keeping the same interface. JavaScript implements all four — and knowing this terminology is critical for any developer interview in India.

02 Declaring a Class

A class is a blueprint for creating objects. The constructor method runs automatically when you create a new instance with the new keyword. It initialises the instance's properties. Methods defined in the class body are added to every instance's prototype — they are shared across all instances, not copied, which is memory-efficient. JavaScript classes (introduced in ES6) are syntactic sugar over the existing prototype-based inheritance system — the underlying mechanism is the same, but the syntax is dramatically cleaner.

```
class UserProfile {
```

Class declaration — capitalise class names by convention

```
  constructor(name, username, followers = 0) {
```

Runs when 'new UserProfile(...)' is called

```
    this.name = name;
```

'this' refers to the new instance being created

```

    this.username = username;
    this.followers = followers;
    this.links = [];
  }
  displayProfile() {
    return `${this.name} (@${this.username}) — ${this.followers} followers`;
  }
  addLink(title, url) {
    this.links.push({ title, url, clicks: 0, isActive: true });
  }
}
const aarya = new UserProfile('Aarya Patel', 'aarya_creates', 1500);
aarya.addLink('Instagram', 'https://instagram.com/aarya_creates');
console.log(aarya.displayProfile());

```

Every new profile starts with an empty links array

Instance method — available on all instances

Method that modifies the instance's data

Create an instance — aarya is an object

Call instance method

'Aarya Patel (@aarya_creates) — 1500 followers'

03 Inheritance with extends & super

CONCEPT — extends & super

The extends keyword makes a class inherit from another. The child class gets all properties and methods of the parent. Inside the child constructor, you must call super() before accessing this — super() calls the parent's constructor to set up inherited properties. Override a parent method in the child by defining a method with the same name — the child's version takes precedence. Call the parent's version of a method explicitly using super.methodName().

```

class CreatorProfile extends UserProfile {
  constructor(name, username, followers, niche) {
    super(name, username, followers);
    this.niche = niche;
    this.isVerified = followers > 10000;
  }
  displayProfile() {
    const base = super.displayProfile();
    return `${base} | ${this.niche} ${this.isVerified ? '' : ''}`;
  }
}
const creator = new CreatorProfile('Aarya Patel', 'aarya_creates', 15000, 'Tech');

```

CreatorProfile inherits everything from UserProfile

Additional parameter: content niche

MUST call super() first — sets up name, username, followers, links

Additional property specific to CreatorProfile

Auto-verify if above threshold

Override parent method — polymorphism

Call parent's version, then extend it

```
console.log(creator.displayProfile());
```

```
'Aarya Patel (@aarya_creates) — 15000 followers |  
Tech'
```

04 Static Methods & Private Fields

```
class UserProfile {  
  #password;  
  
  constructor(name, username, password) {  
    this.name = name;  
    this.username = username;  
    this.#password = password;  
  }  
  checkPassword(input) {  
    return input === this.#password;  
  }  
  static createGuest() {  
    return new UserProfile('Guest', 'guest_' + Date.now(), 'temp123');  
  }  
}  
const guest = UserProfile.createGuest();
```

prefix = private field — not accessible outside the class (ES2022)

Set private field — only accessible inside the class

Method provides controlled access to private field

Static method — called on the CLASS, not an instance

Static method called on class — no 'new UserProfile()' first

TRY THIS LIVE

Build a complete LinkManager class with: a constructor that initialises an empty links array and loads from localStorage; an addLink(title, url) method that validates, adds, and saves; a removeLink(title) method that filters out the link and saves; a getActiveLinks() method that returns only links where isActive is true; and a static fromLocalStorage() method that returns a new LinkManager with saved data pre-loaded. Use this class to replace all the scattered link functions in your app.

PRO TIP — OOP vs Functional in React

React class components (written before 2019) used OOP heavily — extends React.Component, this.state, this.setState. Modern React uses functional components with hooks — a more functional programming style. However, you will still encounter OOP in: custom Error classes, service layer objects (API clients), utility classes, and any legacy codebase. Understanding both OOP and functional programming makes you adaptable to any project.

CASE STUDY — Razorpay (Bengaluru)

Context — Razorpay is India's leading payment gateway, powering over 8 million businesses including numerous Gujarat-based e-commerce and D2C brands.

Challenge — Razorpay's JavaScript SDK needed to handle multiple payment methods (UPI, card, netbanking, wallet, EMI) with shared base logic but different validation, display, and processing behaviour for each type.

What they did — Engineers modelled each payment method as a class extending a common `PaymentMethod` base class. Shared logic (amount validation, currency formatting, error handling) lives in the parent. Method-specific overrides (UPI VPA validation vs card number validation) live in each subclass — a textbook use of inheritance and polymorphism.

Result — Adding a new payment method required creating one new subclass — not modifying existing code. The SDK grew from 3 to 9 payment methods with minimal regression risk.

Takeaway — Inheritance lets you add new behaviour without rewriting existing code — the Open/Closed Principle, a cornerstone of maintainable software.

Source: Razorpay Engineering Blog — SDK Architecture, 2022

WHERE TO GO NEXT

You now understand the OOP concepts that power large-scale JavaScript applications — and that formed the basis of React before hooks existed. Both paradigms appear in every professional codebase.

Explore deeper — MDN Docs: 'Classes' —

developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Classes — covers getters, setters, private methods, and static fields

Try on your own — Add getter and setter methods to your `UserProfile` class: a `get followerCount()` getter that returns followers formatted as '1.5K' or '1.2M', and a `set followerCount(value)` setter that validates the number is non-negative before setting.

Connect the dots — Session 16 introduces advanced array methods — `map`, `filter`, `reduce` — which are the functional-programming counterparts to the OOP patterns you learned today. Both styles coexist in real codebases.

Going further — Read about JavaScript's actual prototype chain — how class syntax is implemented under the hood using `Object.create()` and prototype delegation. Understanding this helps you debug class-related issues and read older pre-ES6 OOP code.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript OOP instructor. Explain the concept of inheritance using a real Indian business example: a parent class 'Employee' and child classes 'Manager' and 'Intern' at an IT company in Ahmedabad. Show constructor and one overridden method. Keep it under 150 words.

SESSION

16

60 MIN · FUNCTIONAL PROGRAMMING

Advanced Array Methods

◆ Learning Objectives

- Transform arrays using `.map()` to produce new arrays of equal length
- Filter arrays using `.filter()` to produce arrays containing only matching elements
- Accumulate arrays into single values using `.reduce()`
- Search arrays using `.find()`, `.findIndex()`, `.some()`, and `.every()`
- Chain multiple array methods to perform multi-step data transformations
- Apply these methods to the Bio-Link App's links array for sorting, filtering, and analytics

01 Why Advanced Array Methods?

The advanced array methods — `map`, `filter`, `reduce`, `find`, `some`, `every` — are the functional programming tools of JavaScript. Each accepts a callback function and applies it to every element in the array, producing a transformed result without mutating the original. These methods are the core of React rendering: every list of components you render in React uses `.map()`. Every feature — show only active links, calculate total clicks, find the most popular link — is one of these methods applied to your data array.

02 `.map()` — Transform Every Element

CONCEPT — `.map()`

`.map()` creates a NEW array by applying a callback to every element of the original. The new array always has the same number of elements as the original — only the values are transformed. The original array is NOT modified. In React, `map()` is used on every render to convert an array of data objects into an array of JSX elements — for example, converting links array into a list of `<LinkCard>` components.

```
const links = [
  { title: 'Instagram', url: 'https://instagram.com/aarya', clicks: 234, isActive: true },
  { title: 'YouTube', url: 'https://youtube.com/@aarya', clicks: 512, isActive: true },
  { title: 'LinkedIn', url: 'https://linkedin.com/in/aarya', clicks: 89, isActive: false }
];

const titles = links.map(link => link.title);    Extract just the title from each link object
console.log(titles);                            ['Instagram', 'YouTube', 'LinkedIn']

const cardHTML = links.map(link =>              Transform each link into an HTML string
`<div class='card'><a href='${link.url}'>${link.title}</a></div>`
```

```
);
```

cardHTML is now an array of HTML strings — same length as links

```
document.querySelector('#container').innerHTML = cardHTML.join('');
```

Join into one string and inject into the DOM

03 .filter() — Keep Only What Matches

`.filter()` creates a new array containing only the elements for which the callback returns true. The original array is unchanged. The returned array is usually shorter — only elements that pass the test are included. In the Bio-Link App, `filter()` powers the 'show only active links' feature. In an e-commerce app like Ajo, every price range filter, category filter, and availability filter is a `.filter()` on the products array.

```
const activeLinks = links.filter(link => link.isActive);
```

Keep only active links — returns 2 of 3

```
console.log(activeLinks.length);
```

2 — LinkedIn (isActive:false) was filtered out

```
const highTraffic = links.filter(link => link.clicks > 200);
```

Links with more than 200 clicks

```
console.log(highTraffic.map(l => l.title));
```

['Instagram', 'YouTube'] — chaining map after filter

```
const removeLink = (links, title) =>
```

filter is the React-safe way to remove from an array

```
  links.filter(link => link.title !== title);
```

Returns NEW array without the removed link

```
const updated = removeLink(links, 'LinkedIn');
```

updated has 2 elements; links is unchanged

04 .reduce() — Accumulate to a Single Value

CONCEPT — .reduce()

`.reduce()` is the most powerful and most intimidating array method. It processes each element and accumulates a running result (the accumulator). The callback receives (accumulator, currentElement) and must return the new accumulator value. The second argument to `reduce()` is the initial value of the accumulator. Use `reduce()` to sum numbers, count occurrences, build objects from arrays, or flatten nested arrays — any transformation that produces a single result from many inputs.

```
// Sum all clicks
```

```
const totalClicks = links.reduce((sum, link) => sum + link.clicks, 0);
```

Initial accumulator = 0. Adds each link's clicks to running sum.

```
console.log(totalClicks);
```

835 (234 + 512 + 89)

```
// Count active vs inactive
```

```
const counts = links.reduce((acc, link) => {
```

Accumulator starts as an object

```
const key = link.isActive ? 'active' : 'inactive';
  acc[key] = (acc[key] || 0) + 1;
  return acc;
}, { active: 0, inactive: 0 });
console.log(counts);
```

Increment the relevant counter
Must return the accumulator
Initial value: object with two counters
{ active: 2, inactive: 1 }

05 .find(), .findIndex(), .some(), .every()

```
// .find() - returns the FIRST matching element (or undefined)
const instaLink = links.find(link => link.title === 'Instagram');
  Returns the Instagram link object — not an array
console.log(instaLink?.clicks);                234 — safe access with optional chaining
// .findIndex() - returns the INDEX of first match (or -1)
const idx = links.findIndex(link => link.title === 'YouTube');
  Returns 1 — YouTube is at index 1
// .some() - true if AT LEAST ONE element passes the test
const hasHighTraffic = links.some(link => link.clicks > 500);
  true — YouTube has 512 clicks
// .every() - true if ALL elements pass the test
const allActive = links.every(link => link.isActive);
  false — LinkedIn is inactive
```

TRY THIS LIVE

Using your Bio-Link App's links array, write five one-liners: (1) Get an array of only the URLs using map. (2) Get only inactive links using filter. (3) Calculate average clicks using reduce. (4) Find the link with title 'GitHub' using find. (5) Check if any link has more than 1000 clicks using some. Chain filter and map in one expression to get titles of active links only.

PRO TIP — Method Chaining for Data Pipelines

Array methods return arrays, so they can be chained: `links.filter(...).map(...).reduce(...)`. Read a chain left to right — filter first narrows the data, map transforms what remains, reduce aggregates the result. This functional pipeline style is extremely readable and is the standard pattern in data-heavy React components. However, chains longer than 3-4 steps are usually a sign to break the logic into named intermediate variables for clarity.

REACT CODEBASES USING `.MAP()` FOR RENDERING**~98%**

Proportion of React component files containing at least one `.map()` call for list rendering — making it arguably the single most-used JavaScript method in front-end development (State of JS survey, 2023)

WHERE TO GO NEXT

You can now perform any data transformation on arrays with clean, readable functional code. Every React component that renders a list is just a `.map()` — which you now fully understand.

Explore deeper — JavaScript.info: 'Array methods' — javascript.info/array-methods — interactive exercises and comprehensive coverage of every array method

Try on your own — Sort your links array by clicks (highest first) using `links.sort((a, b) => b.clicks - a.clicks)`. Then use `filter` + `map` to render only active links sorted by clicks. Build a simple 'top links' leaderboard in the DOM.

Connect the dots — Session 17 covers event delegation and dynamic UI — you will use `filter` and `map` to re-render the links list every time a link is added, removed, or toggled, implementing a proper reactive update cycle.

Going further — Research `Array.from()` and the iterator protocol — how to convert `NodeLists`, `Sets`, `Maps`, and custom iterable objects into arrays to use all of today's methods on them.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript tutor. Explain `.map()`, `.filter()`, and `.reduce()` in simple Hinglish using a practical Indian business example: a kirana store owner who (1) maps prices to include GST, (2) filters out out-of-stock items, (3) reduces the cart to a total bill. Show one line of code for each. Keep it under 150 words.

SESSION

17

60 MIN · BROWSER & DOM

Event Delegation & Dynamic UI Updates

◆ Learning Objectives

- Explain the difference between event bubbling and event capturing in the DOM
- Implement event delegation to handle events on dynamic child elements efficiently
- Use `event.target` to identify exactly which element triggered a delegated event
- Re-render the entire UI from a data array (single source of truth pattern)
- Add, remove, and toggle links using data-driven UI updates
- Implement a delegated click handler to delete link cards without attaching per-card listeners

01 The Problem — New Elements Miss Event Listeners

You attached event listeners to link cards in Session 07. But there is a critical problem: event listeners attached at page load only apply to elements that exist at that moment. When a user adds a new link card dynamically, that new card has no listeners — clicking its Delete button does nothing. You could re-attach listeners after every add, but that quickly creates duplicate listeners on old cards. Event delegation solves this elegantly: instead of listening on each child card, you listen on the stable parent container — and use event bubbling to catch all child events in one place.

CONCEPT — Event Bubbling Revisited

When a user clicks the Delete button inside a link card inside the `#links-container`, three elements receive the click event in sequence: first the Delete button, then the link card div, then the `#links-container`. This upward propagation is bubbling. Event delegation exploits this: attach ONE listener to `#links-container` and inspect `event.target` to find out exactly which button was clicked. The listener fires for every child — including dynamically added ones — because the parent container always exists.

02 Implementing Event Delegation

```
const container = document.querySelector('#links-container');
```

Stable parent — exists from page load

```
container.addEventListener('click', (event) => {
```

ONE listener catches ALL clicks inside the container

```
const deleteBtn = event.target.closest('[data-action="delete"]');
```

`closest()` finds the nearest ancestor matching the selector

```
const editBtn = event.target.closest('[data-action="edit"]');
```

Same pattern for edit buttons

```
if (deleteBtn) {
```

```
  const title = deleteBtn.dataset.title;
```

Read the link title from a data attribute on the button

```
links = links.filter(link => link.title !== title);
```

Remove from data array using filter (non-mutating)

```
  renderLinks(links);
  saveLinks(links);
}
});
```

Re-render the UI from the updated data

Persist to localStorage

Works for ALL current and future cards — no per-card listeners needed

03 data-* Attributes — Attaching Data to Elements

HTML data-* attributes let you store custom data on any element, accessible in JavaScript via the element.dataset object. Instead of searching the DOM to figure out which link a button belongs to, you embed the link's identifier directly on the button as a data attribute when you create the card. This is the standard pattern for building interactive lists — it avoids complex DOM traversal and keeps your event handler simple and readable.

```
const createLinkCard = (link) => {
  const card = document.createElement('div');
  card.className = 'link-card';
  card.dataset.id = link.title;
  card.innerHTML = `
    <a href='${link.url}' target='_blank'>${link.title}</a>
    <span class='clicks'>${link.clicks} clicks</span>
    <button data-action='delete' data-title='${link.title}'></button>
    <button data-action='toggle' data-title='${link.title}'>
      ${link.isActive ? 'Hide' : 'Show'}
    </button>
  `;
  return card;
};
```

Updated card creator — uses data attributes

Store identifier on the card itself

Template literal for inner HTML

data-action and data-title let the delegated handler identify this button

04 Single Source of Truth — Rendering from Data

CONCEPT — Data as the Source of Truth

The most important architectural shift in this session: never modify the DOM directly to reflect application state. Instead, modify the DATA (your links array) first, then re-render the entire UI from that data. This pattern — render(data) — is exactly how React, Vue, and all modern UI frameworks work. Your data is always accurate; your UI is always a reflection of your data. This eliminates an entire class of bugs where the DOM shows something different from what your data actually contains.

```
const renderLinks = (linksArray) => {
  const container = document.querySelector('#links-container');
  container.innerHTML = '';
  if (linksArray.length === 0) {
    container.innerHTML = '<p class="empty">No links yet. Add your first link!</p>';
    return;
  }
  linksArray.forEach(link => {
    container.appendChild(createLinkCard(link));
  });
};
// Any state change follows the same pattern:
// 1) Update data array 2) renderLinks(data) 3) saveLinks(data)
```

Single render function — replaces entire UI

Clear current UI completely

Handle empty state

Build and insert each card from data

Three steps for every user action

TRY THIS LIVE

Complete the delegated click handler to also handle the 'toggle' action: when the Toggle button is clicked, find the link in the array by title, flip its `isActive` property using the spread operator (`{ ...link, isActive: !link.isActive }`), rebuild the array using `map`, re-render, and save. Also add a 'Sort by Clicks' button outside the container that sorts the links array by clicks and re-renders. Notice: you never touch individual DOM nodes — only the data changes.

05 Capturing Phase & stopPropagation

CONCEPT — Capturing vs Bubbling

Events travel in three phases: capture (document -> target), target, then bubble (target -> document). `addEventListener`'s optional third argument controls which phase you listen on: `{capture: true}` fires during the downward capture phase, `false` (default) fires during the upward bubble phase. You can stop either phase with `event.stopPropagation()`. In practice, 99% of handlers use the bubble phase. The capture phase is occasionally used for global keyboard shortcuts or drag-and-drop systems that need to intercept events before they reach child elements.

PRO TIP — Use `closest()` Instead of Checking target Directly

When a button contains child elements (like an SVG icon), `event.target` might be the icon inside the button, not the button itself. Always use `event.target.closest("[data-action]")` instead of checking `event.target` directly — `closest()` walks up the DOM tree and returns the first matching ancestor, ensuring you find the button regardless of which child element was actually clicked. This eliminates an entire category of delegation bugs.

WHERE TO GO NEXT

Your Bio-Link App now uses professional data-driven rendering with a single delegated event handler. This is the mental model React is built on — you have essentially built a tiny custom React in vanilla JS.

Explore deeper — MDN Docs: 'Element.closest()' —

developer.mozilla.org/en-US/docs/Web/API/Element/closest — essential companion to event delegation

Try on your own — Add a search input above the links container. On every input event, filter the links array by title (case-insensitive) and call `renderLinks()` with the filtered subset. The live search should update the displayed cards as the user types — without adding any new listeners.

Connect the dots — Session 18 is the final technical session before the mini-project: debugging tools that will help you find and fix issues in the code you have built across all previous sessions.

Going further — Study the Virtual DOM — how React creates an in-memory representation of the UI, diffs it against the previous version, and applies only the minimum necessary real DOM changes. This is React's answer to the performance cost of `innerHTML = "` on every re-render.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript teacher. Explain event delegation in simple Hinglish. Use the analogy of a single manager at a call centre who handles all incoming calls (events) from different customers (child elements) rather than each customer being assigned their own dedicated agent. Keep it under 150 words.

SESSION

18

60 MIN · DEVELOPER WORKFLOW

Debugging & Developer Tools

◆ Learning Objectives

- Use all console methods effectively: log, warn, error, table, group, time
- Set breakpoints in the Sources tab and step through code line by line
- Inspect and modify variables in scope during a paused debugging session
- Read and interpret a JavaScript stack trace to locate the source of an error
- Use the Network tab to inspect API requests, responses, and timing
- Debug the addLink() function and fix at least two intentionally introduced bugs

01 Why Debugging is a Core Skill

Professional developers spend 30-50% of their time debugging — finding and fixing problems in code. Beginner developers rely only on `console.log()`; professional developers use the full browser DevTools suite. Mastering debugging reduces the time to fix a bug from hours to minutes. It also reduces panic: when you know exactly how to inspect any running program's state, errors stop being mysterious and become simply the next puzzle to solve. Every session from here — React, Firebase, Redux — will produce bugs. The tools you learn today will serve you throughout your entire career.

02 console Methods — Beyond `console.log()`

The console object has over a dozen useful methods. Most developers only know `console.log()`, leaving powerful diagnostic tools unused. `console.warn()` and `console.error()` produce yellow and red highlighted messages with stack traces — ideal for signalling problems. `console.table()` renders an array of objects as a formatted table — perfect for inspecting your links array. `console.group()` groups related logs under a collapsible header. `console.time()` measures how long a block of code takes — essential for performance debugging.

```
console.log('Regular message:', links);           Standard log — shows the object/array as interactive
console.warn('No links found — loading defaults');
  Yellow highlight — caution, not a crash
console.error('localStorage read failed:', error);
  Red highlight — includes stack trace
console.table(links);                             Renders the links array as a sortable table — excellent
                                                    for arrays of objects
console.group('Link Add Operation');               Start a collapsible group in console
  console.log('Input title:', title);
  console.log('Input url:', url);
console.log('Links after add:', links.length);
console.groupEnd();                               End the group
```

```
console.time('renderLinks');
renderLinks(links);
console.timeEnd('renderLinks');
```

Start a timer

Prints: 'renderLinks: 2.3ms' — measure render performance

03 The Sources Tab — Breakpoints & Step-Through

CONCEPT — Breakpoints

A breakpoint is a marker you set on a line in your source code that tells the browser: 'Pause execution here.' When the browser hits that line, it freezes the program — everything stops. You can then inspect every variable in scope, step to the next line, step into a function call, or resume execution. This lets you see exactly what your program knows at any moment. It is the single most powerful debugging technique and is universally used by professional developers — yet most beginners never use it.

Action	Keyboard Shortcut	What It Does
Resume	F8	Continue running until next breakpoint
Step Over	F10	Execute current line, move to next — don't enter functions
Step Into	F11	Enter the function being called on current line
Step Out	Shift+F11	Finish current function, return to caller
Deactivate All	Ctrl+F8	Temporarily disable all breakpoints

04 Reading a Stack Trace

A stack trace is the error message the console shows when an uncaught error occurs. It lists every function call that led to the crash, in reverse order — the most recent call first. Reading from bottom to top shows the chain of events: what called what. The file name and line number beside each function are clickable — clicking them opens that exact line in the Sources tab. Stack traces feel overwhelming at first, but they are the most direct path to finding where and why something broke.

```
TypeError: Cannot read properties of null (reading 'title')
```

Error type and message — what went wrong

```
at renderCard (dom.js:23:18)
```

Most recent call — the line that crashed: dom.js line 23, column 18

```
at renderLinks (dom.js:11:5)
```

renderCard was called from here

```
at addLink (main.js:44:3)
```

renderLinks was called from here

```
at HTMLButtonElement.<anonymous> (main.js:18:5)
```

addLink was triggered by a click event handler

```
// Reading strategy: start at the TOP (crash site)
```

```
// dom.js line 23: link.title — 'link' must be null here
```

Root cause: a null link object was passed to renderCard

05 The Network Tab — Inspecting API Calls

Every `fetch()` call your app makes appears in the Network tab (F12 -> Network). Click any request to see: the full URL, HTTP method, request headers, request body (for POST), response status code, response headers, and the full response body. This tab is invaluable for debugging API issues — you can immediately see whether the request was made, whether it reached the server, what the server returned, and how long it took. Filter by 'Fetch/XHR' to see only API calls, filtering out CSS, image, and font requests.

TRY THIS LIVE

Open your Bio-Link App with DevTools open on the Sources tab. Set a breakpoint on the first line of your `addLink` function. Fill in the form and click Add Link — execution should pause at your breakpoint. In the right panel, expand the 'Scope' section and find the `title` and `url` variables. Check their values match what you typed. Step through each line with F10 and watch the `links` array grow in the Scope panel. This is the single most important debugging skill you can build.

PRO TIP — `debugger;` Statement

You can trigger a breakpoint directly from JavaScript code by writing `debugger;` on any line. When DevTools is open and execution reaches `debugger;`, it pauses exactly as if you had set a breakpoint in the Sources tab. This is useful when you cannot easily click a line in Sources (e.g., a deeply nested file in a bundled app) or when you want to commit the breakpoint to your code temporarily during a debugging session. Remove all `debugger;` statements before committing to version control.

CONCEPT — Common Bug Patterns to Watch For

The most common JavaScript bugs fall into predictable patterns: (1) undefined or null access — always check a value exists before accessing properties. (2) Async timing — code runs before a Promise resolves. (3) Wrong this context — methods called incorrectly lose their 'this'. (4) Off-by-one in loops — using `<` vs `<=`. (5) Mutating shared state — two functions modify the same array unexpectedly. Recognising these patterns

makes you dramatically faster at finding bugs.

WHERE TO GO NEXT

You now have the complete professional debugging toolkit. When something breaks in React, Next.js, or any future project, the DevTools approach here applies without change.

Explore deeper — Chrome DevTools official docs: developer.chrome.com/docs/devtools — comprehensive guide to every panel including Performance, Memory, and Application tabs

Try on your own — Intentionally introduce a bug into your Bio-Link App — for example, remove the null check in `loadLinks()` — and use the Sources tab breakpoint (not `console.log`) to find and identify the exact line and variable causing the error. Practice debugging three different bugs: one `TypeError`, one logic error, one async timing issue.

Connect the dots — Session 19 begins the mini-project integration phase — combining all 18 sessions' worth of concepts into one cohesive, production-ready web application.

Going further — Look into VS Code's built-in JavaScript debugger — you can set breakpoints and step through code directly inside VS Code without switching to the browser. It connects via the browser's remote debugging protocol and is the preferred workflow for Node.js debugging.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a senior JavaScript developer. Explain how to read a stack trace in simple Hinglish. Use the analogy of investigating a chain of events in an office: which peon gave which file to which manager, tracing back to where the mistake originally happened. Keep it under 150 words.

SESSION

19

60 MIN · PROJECT

Mini-Project Preparation & Integration

◆ Learning Objectives

- Map every feature of the Social Bio Link Builder to the JavaScript concepts learned
- Design the complete data flow: Form input -> Validation -> Storage -> DOM render
- Organise project files into a clean module structure before writing integration code
- Identify and resolve dependency order between modules
- Plan and sketch the app architecture on paper before coding
- Begin integrating all module files into a functioning end-to-end prototype

01 What We Are Building

The Social Bio Link Builder is a single-page web application that lets any content creator in India maintain a personal page of social media links — similar to Linktree or Biosite, but built entirely by you from scratch. The finished app will let users add, edit, delete, and reorder social links; persist all data across page reloads; toggle individual links between visible and hidden; apply a light/dark mode; and validate all inputs before accepting them. Every feature maps directly to a session from this module.

App Feature	JavaScript Concept	Session
Display profile name & bio	DOM manipulation, <code>textContent</code>	06
Add a new social link	Events, form handling, objects	07, 08
Validate URL and title input	Error handling, custom errors, <code>throw</code>	13
Save links across reloads	JSON, <code>localStorage</code>	09
Render links from data array	<code>map()</code> , event delegation, <code>data-*</code>	16, 17
Delete a link	<code>filter()</code> , re-render from data	16, 17
Toggle link visibility	<code>map()</code> with spread, re-render	10, 16
Dark mode toggle	<code>classList.toggle()</code> , <code>localStorage</code> preference	06, 09
Fetch mock user on load	<code>fetch()</code> , <code>async/await</code> , loading state	12
Modular file structure	ES Modules, <code>import/export</code>	14

02 Project File Structure

CONCEPT — Separation of Concerns

Separation of concerns means each file is responsible for exactly one thing. When a bug occurs in storage, you look only in storage.js. When the UI breaks, you look only in dom.js. This dramatically reduces the mental load of debugging and the risk of changes in one area breaking another. The file structure below is the industry standard for small vanilla JS projects — and directly maps to how React projects are organised into components, hooks, and service files.

<code>bio-link-app/</code>	Root project folder
<code>index.html</code>	Entry point — HTML skeleton, imports main.js as a module
<code>style.css</code>	All CSS — cards, dark mode, responsive layout
<code>js/</code>	All JavaScript files — separated by responsibility
<code>config.js</code>	Constants: APP_NAME, STORAGE_KEY, DEFAULT_THEME
<code>utils.js</code>	Pure functions: validateURL, generateId, formatClicks
<code>storage.js</code>	localStorage: saveLinks, loadLinks, saveTheme, loadTheme
<code>api.js</code>	Fetch calls: fetchMockUser, postLink
<code>dom.js</code>	DOM: createLinkCard, renderLinks, showError, setLoading
<code>main.js</code>	Entry point: imports all modules, wires events, initialises app

03 Designing the Data Flow

Before writing code, every feature should be mapped as a data flow: what input triggers it, what data changes, what gets saved, and what UI update results. This planning step prevents half-built features and tangled code. The diagram below is what you should sketch on paper or a whiteboard before integration — professional developers call this a 'flow diagram' or 'sequence diagram'. It is the equivalent of a blueprint before construction.

CONCEPT — Add Link Data Flow

User submits form -> main.js form submit handler fires -> reads title and url from inputs -> calls `utils.validateURL(url)` (throws `ValidationError` if invalid) -> caught by try/catch, shown in UI -> if valid: creates link object with `generateId()` -> appends to links array using spread -> calls `storage.saveLinks(links)` -> calls `dom.renderLinks(links)` -> clears form inputs. Every step is one function from one module. No step knows about the others — main.js is the only file that imports multiple modules and wires them.

04 Initialisation Sequence

```
// main.js — the complete app initialisation sequence
import { loadLinks, saveLinks } from './storage.js';
import { validateURL, generateId } from './utils.js';
import { renderLinks, showError, setLoading } from './dom.js';
import { fetchMockUser } from './api.js';

let links = []; // App state — single source of truth

document.addEventListener('DOMContentLoaded', async () => {
  // Wait for DOM to be ready
  links = loadLinks(); // Step 1: Load saved links from localStorage
  renderLinks(links); // Step 2: Render them immediately
  const user = await fetchMockUser(); // Step 3: Fetch profile data (async)
  if (user) updateProfileSection(user); // Step 4: Update profile display if fetch succeeded
  wireEvents(); // Step 5: Attach all event listeners
});
```

TRY THIS LIVE

Create the full folder structure above. Create each JS file and add an empty export comment at the top. In index.html, add `<script type='module' src='js/main.js'></script>`. In main.js, import from each module file. Open the page — if no errors appear in the console, your module graph is wired correctly. Then copy functions from your earlier single-file app into their correct new module homes. This refactoring is the most important coding exercise of the module.

05 Whiteboard Architecture Exercise

Before writing the `wireEvents()` function, sketch every user interaction as a box on paper: click Add Link button, click Delete button, click Toggle button, click Dark Mode toggle, type in search box. For each box, draw arrows showing what function is called, which module it lives in, what state changes, and what re-render follows. This exercise makes the actual coding step straightforward because every decision is already made. Senior developers at companies like Freshworks and CRED do exactly this before starting a new feature — on a whiteboard, not in code.

PRO TIP — Build Vertically, Not Horizontally

A common mistake is to build all files partially before any single feature works. Instead, build one complete vertical feature first: Add Link. Get the form -> validate -> add to array -> render -> save flow working end-to-end before touching Delete or Toggle. A working feature is more valuable than five half-built features. This vertical slice approach is used in Agile sprints at every software company in India.

“Code is read far more than it is written. Organise your modules so any developer — including your future self — can understand which file to open within 10 seconds.”

WHERE TO GO NEXT

Your project is planned, structured, and partially wired. Session 20 completes the integration, tests every feature, and polishes the final UI for submission.

Explore deeper — Excalidraw (excalidraw.com) — a free, instant browser-based whiteboard for drawing flow diagrams and architecture sketches — used widely by developers across the Indian startup ecosystem

Try on your own — Write the complete `wireEvents()` function in `main.js` that attaches all event listeners: form submit (add link), delegated container click (delete, toggle), dark mode button click, and search input handler. Use event delegation for the container. Test each listener independently before moving on.

Connect the dots — Session 20 is the final session: integration testing, common bug fixes, UI polish, and the light/dark mode toggle — completing the Social Bio Link Builder as a fully working, shippable project.

Going further — Research 'Vite' — a blazing-fast modern build tool that eliminates the need for `type='module'` scripts and enables real npm package imports (like Axios, Lodash) in browser-based JS without a server. React projects in Module 2 will be created with Vite.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a software architect. In simple Hinglish, explain why separating code into modules (like separate departments in a company) is better than writing everything in one file. Give one real consequence of bad organisation (e.g. one change breaks three unrelated features). Keep it under 150 words.

SESSION

20

60 MIN · PROJECT

Integration Testing & Polish

◆ Learning Objectives

- Perform manual QA testing by systematically testing every user interaction
- Identify and fix the five most common Bio-Link App bugs (null values, duplicate adds, empty state)
- Add a light/dark mode toggle with preference saved to localStorage
- Apply basic UI validation patterns: real-time character count, URL format indicator
- Conduct a final run-through of the complete Social Bio Link Builder
- Prepare to explain any part of the codebase in a technical interview or review

01 What Integration Testing Means

Unit testing checks individual functions in isolation. Integration testing checks that all the parts work correctly together as a system — which is what you are doing now. Your modules (utils, storage, dom, api, main) each work in isolation, but integration testing reveals problems at the boundaries: does the data that `utils.validateURL()` passes match what `storage.saveLinks()` expects? Does the link object `dom.createLinkCard()` receives have all the properties it needs? Systematic manual QA is the form of integration testing every developer performs before declaring a feature complete.

02 Manual QA Checklist

Test Case	Expected Result	Common Bug If It Fails
Add a valid link (title + https:// URL)	Card appears, data saved to localStorage	<code>renderLinks()</code> not called after push
Add with empty title	<code>ValidationError</code> shown, no card added	<code>validateURL</code> called but not <code>validateTitle</code>
Add with http:// URL (not https)	Error shown: 'Must start with https://'	<code>validateURL</code> regex not checking protocol
Add the same title twice	Second add blocked OR both added with unique IDs	No duplicate check — <code>find()</code> not used
Delete a link	Card removed, localStorage updated, count correct	<code>filter()</code> mutates original instead of reassigning
Toggle link to hidden	Card shows 'Hidden' label, <code>isActive</code> flips	Spread not used — mutation bug

Test Case	Expected Result	Common Bug If It Fails
Reload the page	All links reappear exactly as left	loadLinks() not called on DOMContentLoaded
Toggle dark mode	Theme flips, preference survives reload	localStorage key typo, theme not applied on load
Add 10+ links	All render, no layout break, localStorage under 5MB	innerHTML performance on large datasets
Click outside all buttons	Nothing happens — no errors	Delegated handler not guarding against null target

03 The Five Most Common Bugs & Their Fixes

WARNING — The Duplicate Add Bug

If a user clicks 'Add Link' twice quickly (or submits the form twice before the UI updates), duplicate cards appear. Fix: (1) Disable the submit button immediately on click and re-enable in finally. (2) Add a duplicate check: if (links.some(l => l.title === title)) throw new ValidationError('Title already exists', 'title'). Both defences should be present — one is UX protection, one is data integrity.

```
// Bug 1: Links not persisting — wrong reassignment
```

```
links.push(newLink);
```

BUG: mutates array but links variable still points to old reference

```
links = [...links, newLink];
```

FIX: reassign links to a new array

```
// Bug 2: Delete removes wrong link — index vs title
```

```
links.splice(index, 1);
```

BUG: indices shift after first deletion — wrong link removed next time

```
links = links.filter(l => l.id !== linkId);
```

FIX: use unique ID, not index — filter creates new array

```
// Bug 3: Dark mode not applied on load
```

```
const savedTheme = localStorage.getItem('theme');
```

```
if (savedTheme === 'dark') document.body.classList.add('dark');
```

FIX: apply saved theme before any render — in the very first line of DOMContentLoaded

```
// Bug 4: Null link passed to renderCard
```

```
links.forEach(link => renderCard(link));
```

BUG: if loadLinks() returns null (corrupt data), forEach crashes

```
const safeLinks = Array.isArray(links) ? links : [];
```

FIX: always validate shape of loaded data before using

04 Light / Dark Mode Toggle

```
// CSS (style.css)
body { background: #ffffff; color: #1a1a1a; transition: background 0.3s, color 0.3s; }
  Default light theme
body.dark { background: #121212; color: #f5f5f5; }
  Dark theme — toggled by adding/removing .dark class
.link-card { background: #f0f0f0; } body.dark .link-card { background: #1e1e1e; }
  Card adapts to both themes

// JavaScript (main.js)
const themeBtn = document.querySelector('#theme-toggle');
themeBtn.addEventListener('click', () => {
  document.body.classList.toggle('dark');      Toggle the class on body
const theme = document.body.classList.contains('dark') ? 'dark' : 'light';
  Read current state
  localStorage.setItem('theme', theme);          Save preference
themeBtn.textContent = theme === 'dark' ? ' Light Mode' : '■ Dark Mode';
  Update button label
});
// Apply saved theme on load (before render)
const savedTheme = localStorage.getItem('theme') ?? 'light';
if (savedTheme === 'dark') document.body.classList.add('dark');
```

05 UI Validation Polish

Good UI validation gives users immediate, helpful feedback — not just an error after they click submit. Real-time validation shows a green checkmark as they type a valid URL, a character count for the title field, and a red border on any invalid field. This is not extra polish — it directly reduces form abandonment. Every major Indian form-based app (Naukri job applications, Ola ride registration, Paytm KYC) uses real-time validation to keep users engaged through the form completion process.

```
const titleInput = document.querySelector('#title-input');
const charCount = document.querySelector('#char-count');
  A <span id='char-count'>0/30</span> next to the input
titleInput.addEventListener('input', () => {      Fires on every keystroke
  const len = titleInput.value.length;
  charCount.textContent = `${len}/30`;              Update character count
charCount.style.color = len > 25 ? 'coral' : '#888';
  Warn when approaching limit
});
const urlInput = document.querySelector('#url-input');
```

```
urlInput.addEventListener('input', () => {  
  const isValid = urlInput.value.startsWith('https://');  
  urlInput.classList.toggle('valid', isValid);  
  Add 'valid' class (green border) when correct  
  urlInput.classList.toggle('invalid', urlInput.value && !isValid);  
  Add 'invalid' class (red border) only when user has typed something wrong  
});
```

TRY THIS LIVE

Run through the full QA checklist from Section 02 — test every single row. For each bug you find, fix it before moving to the next test. Keep DevTools open throughout: Network tab visible while testing the fetch on load, Console tab visible for errors, Application tab open to inspect localStorage after every add/delete. Document any bug you fix: what the symptom was, what caused it, and what the fix was. This documentation habit is expected in professional teams.

PRO TIP — The Code Review Mindset

Before finalising any project, read every line of code as if you are a stranger seeing it for the first time. Does each function name describe exactly what it does? Is every magic number (like 30 for the title limit) extracted into a named constant? Is every catch block handling the error, not just silently swallowing it? Is every fetch call protected against null responses? These are the exact questions a senior developer will ask in a code review at a company like Juspay, Meesho, or CRED.

“Shipping code that works under all conditions — including edge cases, empty states, and network failures — is the difference between a developer and a professional developer.”

COST OF FIXING A BUG FOUND IN PRODUCTION VS IN TESTING

10-100x

Industry research consistently shows bugs found and fixed during testing cost dramatically less than bugs discovered after release (IBM Systems Sciences Institute; widely cited in software engineering)

WHERE TO GO NEXT

The Social Bio Link Builder is complete. You have built a real, working web application using every core JavaScript concept — from variables to async, from classes to modules. This is your first portfolio piece.

Explore deeper — GitHub Pages (pages.github.com) — deploy your Bio-Link App for free in 5 minutes. Create a GitHub repository, push your project, enable Pages from the repo settings. Share the live URL.

Try on your own — Add one final feature of your choice to the app: (a) drag-to-reorder links using HTML5 drag-and-drop events, (b) an 'analytics' view showing total and per-link click counts using a simple bar chart with CSS widths, or (c) an 'export as JSON' button that downloads the links array as a .json file.

Connect the dots — Module 2 begins React Fundamentals. On Day 1, you will immediately recognise: JSX is template literals, props are function parameters, state is your links array, useEffect is async/DOMContentLoaded — because you have already built all of these in vanilla JS.

Going further — Now that you have a working project, set up ESLint (eslint.org) — a static analysis tool that automatically finds potential bugs and style violations in your JavaScript. It is used in every professional JavaScript codebase in India and worldwide.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a JavaScript senior developer conducting a code review. In simple Hinglish, list five questions you would ask a junior developer about their Bio-Link App code to verify it is production-ready. Cover: error handling, edge cases, performance, security (XSS), and code readability. Keep it under 150 words.

REVIEW



INTERVIEW PREP · MODULE 1

Interview Questions

◆ JavaScript Essentials And Advanced

Rehearse these questions out loud, aiming for 60-second answers. The hint is a nudge to guide your thinking; the answer is a model to compare against after you've tried. Master these topics to ace JavaScript interviews.

1

What is JavaScript, and how does it differ when used on the client side versus the server side?

Think about where the code runs and what it can access.

MODEL ANSWER

JavaScript is a programming language that runs in browsers (client-side) to make web pages interactive, while server-side JavaScript (Node.js) runs on servers to build backend services. Client-side JS accesses the DOM and user interactions, whereas server-side JS handles data and business logic.

2

How do you include JavaScript in an HTML page so that it interacts with the web page's content?

Which HTML tag is used, and where is it usually placed?

MODEL ANSWER

You include JavaScript in HTML using the `<script>` tag. It can be placed in the `<head>` section or at the end of the `<body>` tag. Placing it at the end of `<body>` is preferred because it allows HTML to load first, improving page performance.

3

If you want to quickly test a line of JavaScript code, where can you do this in your browser?

Most browsers have a built-in tool for developers.

MODEL ANSWER

You can test JavaScript code directly in the browser's Developer Console, accessible via F12 or Ctrl+Shift+I. The console lets you run code snippets instantly and see results, making it ideal for quick testing and debugging.

4

Explain the difference between `<code>var</code>`, **`<code>let</code>`**, and **`<code>const</code>`** in JavaScript.

MODEL ANSWER

var, let, and const are three ways to declare variables. var has function scope and can be redeclared; let has block scope and cannot be redeclared; const is block-scoped, cannot be redeclared, and cannot be reassigned. Use const by default, let when reassignment is needed, and avoid var in modern code.

5

What is the difference between primitive and reference data types in JavaScript?

Think about how values are stored and copied.

MODEL ANSWER

Primitive types (number, string, boolean, null, undefined) are immutable and stored directly as values, while reference types (objects, arrays) are stored as references in memory. When you copy a primitive, you get an independent value; copying a reference creates two pointers to the same object.

6

Give an example of dynamic typing in JavaScript.

MODEL ANSWER

Dynamic typing means variables can change their data type at runtime. For example: let x = 10 (number), then x = 'hello' (string), then x = true (boolean) are all valid. The type is determined by the assigned value, not declared upfront.

7

What will be printed to the console?

```
let followers = 1500;
if (followers > 1000) {
  console.log("Verified Creator");
} else {
  console.log("Regular User");
}
```

Check the value of the variable and the condition.

MODEL ANSWER

The condition followers > 1000 is true (1500 > 1000), so the console will print 'Verified Creator'. This demonstrates how comparison operators evaluate expressions and trigger conditional branches based on the result.

8

Explain the difference between `==` and `===` in JavaScript.

MODEL ANSWER

`==` performs loose comparison and converts types if needed (e.g., '5' == 5 is true), while `===` performs strict comparison without type conversion (e.g., '5' === 5 is false). Always use `===` in JavaScript to avoid unexpected type coercion bugs.

9

What is a 'truthy' value in JavaScript? Give one example.

Not all values need to be true to behave like true in conditions.

MODEL ANSWER

A truthy value is any value that evaluates to true in a boolean context. Examples include non-zero numbers (5, -1), non-empty strings ('hello'), objects, and arrays. Only falsy values (0, "", null, undefined, false, NaN) evaluate to false.

10

Write a `for` loop that prints each item in this array to the console: `let links = ['Instagram', 'Twitter', 'LinkedIn'];`

MODEL ANSWER

Use a for loop: `for (let i = 0; i < links.length; i++) { console.log(links[i]); }`. This iterates through the array indices, accessing each element and printing 'Instagram', 'Twitter', and 'LinkedIn' to the console.

11

What is the main difference between a `for` loop and a `forEach` loop when working with arrays?

MODEL ANSWER

A for loop iterates using indices and requires manual index management, while `forEach` passes each element directly to a callback function without exposing the index (unless needed as a parameter). `forEach` is more readable for simple iterations, but for loops offer more control.

12

Write a function declaration called `greetUser` that takes a `name` parameter and returns `'Hello, [name]!'`.

MODEL ANSWER

`function greetUser(name) { return 'Hello, ' + name + '!'; }` or using template literals: `return `Hello, ${name}!``. This declares a named function that accepts a name parameter and returns a personalized greeting string.

13

What is the difference between a function declaration and a function expression in JavaScript?

Consider how each can be used before or after its definition.

MODEL ANSWER

A function declaration (`function greet() {}`) is hoisted and can be called before it's defined in code. A function expression (`const greet = function() {}`) is not hoisted and must be defined before use. Arrow functions (`const greet = () => {}`) are also expressions and not hoisted.

14

How would you create a button element in JavaScript and set its label and background color using a function?

Think about using `document.createElement` and setting properties.

MODEL ANSWER

Use `document.createElement('button')` to create the element, set `textContent` for the label, and apply styles via the `style` property: `btn.style.backgroundColor = 'blue'; btn.textContent = 'Click Me'; document.body.appendChild(btn)`. This demonstrates DOM manipulation and property setting.

15

How do you select an element by its ID in the DOM, and how do you change its text content?

There are specific DOM methods for selection and updating.

MODEL ANSWER

Use `document.getElementById('myId')` to select the element by its ID, then update its text with `element.textContent = 'new text'` or `element.innerHTML = '<b>new text</b>'`. `textContent` is safer for plain text; `innerHTML` allows HTML content.

16

Describe how you would add a new link to a list on a web page when a user submits a form.

MODEL ANSWER

Listen for the form's submit event with `addEventListener('submit', (e) => { e.preventDefault(); })`. Create a new link element using `createElement('li')` or `innerHTML`. Append it to the list with `appendChild()`. This shows event handling and dynamic DOM updates.

17

What is the difference between `onclick` and `addEventListener` for handling events in JavaScript?

MODEL ANSWER

`onclick` is an inline event handler attribute (`onclick='doSomething()'`), while `addEventListener` is a method that registers event listeners with `addEventListener('click', doSomething)`. `addEventListener` allows multiple listeners on the same event, is more flexible, and is the modern approach.

18

Explain what the event object is and how you might use it in an event handler.

It provides details about the event that occurred.

MODEL ANSWER

The event object contains information about what happened: target (the element that triggered the event), type (the event type), key (for keyboard events), clientX/clientY (for mouse position), and methods like `preventDefault()`. Access it as the first parameter in the handler: `function(event) {}`.

19

Write a code snippet that adds a click event listener to a button with the ID `addLinkBtn` and logs 'Button clicked!' when the button is pressed.

MODEL ANSWER

`document.getElementById('addLinkBtn').addEventListener('click', function() { console.log('Button clicked!'); });`. This attaches a click listener to the button that logs a message to the console each time the button is pressed.

20

How would you represent a collection of social links, where each link has a title and a URL, using an array of objects in JavaScript?

Think about using curly braces inside an array.

MODEL ANSWER

Use an array of objects: `let links = [{title: 'Instagram', url: 'instagram.com'}, {title: 'Twitter', url: 'twitter.com'}]`. This represents each link as an object with properties, allowing you to store and access complex data. Loop through with `forEach` or `map` to display each link.

21

What is the purpose of `JSON.stringify()` and `JSON.parse()` in JavaScript?

Consider how data is stored or sent as text.

MODEL ANSWER

`JSON.stringify()` converts JavaScript objects to JSON strings for storage or transmission; `JSON.parse()` converts JSON strings back to JavaScript objects. For example: `JSON.stringify({name: 'Ali'})` becomes `'{"name":"Ali"}'`, and parsing it reverses the process.

22

How would you save a user's list of links to `localStorage` and retrieve it later?

Remember, `localStorage` only stores strings.

MODEL ANSWER

Save with `localStorage.setItem('links', JSON.stringify(linksArray))`. Retrieve with `let saved = JSON.parse(localStorage.getItem('links'))`. Always use `stringify` before saving (since `localStorage` only stores strings) and `parse` when retrieving to convert back to objects.

23

What are template literals in JavaScript, and how do they differ from traditional string concatenation?

Look for a special character used to enclose them.

MODEL ANSWER

Template literals use backticks (``) instead of quotes and allow embedded expressions with `${variable}`. Example: ``Hello ${name}`` produces `'Hello Ali'` when `name='Ali'`. They preserve formatting and are more readable than string concatenation.

24

Show how you would use destructuring to extract the `title` and `url` from this object: `let link = { title: 'GitHub', url: 'https://github.com' }`.

MODEL ANSWER

`const {title, url} = link` extracts `title` and `url` from a `link` object into separate variables. This is cleaner than `link.title` and `link.url`. It works with arrays too: `const [first, second] = array` extracts the first two elements.

25

Explain the role of the spread operator (`...`) in copying or merging arrays.

MODEL ANSWER

The spread operator (...) copies array/object contents. Example: `const newArray = [...oldArray, newItem]` creates a new array with all old items plus `newItem`. For objects: `const newObj = {...oldObj, newProp: 'value'}` merges properties.

26

What is the difference between blocking and non-blocking code in JavaScript?

Think about how code execution waits or continues.

MODEL ANSWER

Blocking code halts execution until a task completes (synchronous), while non-blocking code starts a task but continues execution without waiting (asynchronous). Non-blocking is essential for long operations like fetching data; blocking would freeze the user interface.

27

What is a callback function? Give a simple example using `setTimeout`.

MODEL ANSWER

A callback function is passed as an argument to another function and executed later. Example: `setTimeout(() => { console.log('Done!'); }, 2000)` passes a callback that runs after 2 seconds. Callbacks enable delayed or event-driven execution.

28

Write a Promise that resolves after 2 seconds with the message 'Done!'.

MODEL ANSWER

`const promise = new Promise((resolve, reject) => { setTimeout(() => { resolve('Done!'); }, 2000); });`. This creates a promise that automatically resolves after 2 seconds. The resolve function signals successful completion with a value.

29

How does `fetch()` work in JavaScript to get data from an API? Briefly describe the steps involved.

Think about the request, response, and parsing process.

MODEL ANSWER

`fetch()` sends an HTTP request to an API endpoint and returns a Promise. You then call `.then()` to handle the response, convert it to JSON with `.json()`, and use the data. Example: `fetch(url).then(res => res.json()).then(data => console.log(data))`. It's the modern way to load data from servers.

30

Show how you would fetch user data from `https://jsonplaceholder.typicode.com/users` and log the result to the console.

MODEL ANSWER

`fetch('https://jsonplaceholder.typicode.com/users').then(res => res.json()).then(data => { console.log(data); data.forEach(user => console.log(user.name)); })`. This retrieves the user list and prints each user's name using the Promises chain pattern.

31

What is the difference between default exports and named exports in ES6 modules?

Consider how each is imported.

MODEL ANSWER

Default exports export one main item per file (export default function `myFunc()` {}), imported with `import myFunc from './file'`. Named exports export multiple items (export function `myFunc()` {}), imported with `import { myFunc } from './file'`. Use default for the primary export, named for utilities.

32

Why is modularizing your JavaScript code (using imports and exports) beneficial for larger applications?

MODEL ANSWER

Modularizing code splits functionality into separate files, making code reusable, maintainable, and testable. It prevents naming conflicts, allows team members to work on different modules in parallel, and improves scalability. A utility function in `utils.js` can be imported and reused across the entire project.

33

How would you move a utility function from one file to another and use it in your main script?

MODEL ANSWER

Create `utils.js` with: `export function createLink(title, url) { ... }`. In `main.js`, import it with: `import { createLink } from './utils.js'`. Then call `createLink()` directly. This shows how to extract reusable logic and share it across files.

34

How do you declare a class in JavaScript? Write a simple `UserProfile` class with a constructor that takes `name` and `followers`.

Use the `class` keyword.

MODEL ANSWER

`class UserProfile { constructor(name, followers) { this.name = name; this.followers = followers; } displayProfile() { console.log(` ${this.name}: ${this.followers}`); } }`. This declares a class with properties set in the constructor and a method to display the profile.

35

What is the purpose of `try...catch` in JavaScript, and when would you use it?

MODEL ANSWER

`try...catch` lets you handle errors gracefully. Code in `try` runs; if an error occurs, `catch` executes. For example: `try { let x = JSON.parse(InvalidJSON); } catch(e) { console.log("Parse failed"); }` prevents crashes and allows recovery.

36

Write a code snippet that throws a custom error if a provided URL does not start with 'http'.

MODEL ANSWER

`if (!url.startsWith('http')) { throw new Error('Invalid URL'); }`. This checks the URL and throws a custom error if it doesn't start with 'http'. The error message describes the problem, and surrounding code can catch and handle it with `try...catch`.

37

How can AI tools like ChatGPT help you when you encounter an error stack trace in your JavaScript code?

Think about explanations, suggestions, and verifying fixes.

MODEL ANSWER

ChatGPT can explain what the error means, suggest fixes, and help verify solutions. You can paste the stack trace and ask 'Why does this error occur?' to understand root causes. AI tools accelerate debugging but you should verify the explanation and solution yourself.

38

What does the `map()` method do on an array? Give an example.

MODEL ANSWER

`map()` transforms each element using a callback and returns a new array. Example: `const urls = links.map(link => link.url)` extracts all URLs into a new array. It's useful for extracting data or transforming values without modifying the original array.

39

How would you use `filter()` to get only the active links (where `isActive: true`) from an array of link objects?

MODEL ANSWER

`const activeLinks = links.filter(link => link.isActive === true)` returns a new array containing only active links. `filter()` keeps elements where the callback returns true, excluding all others. Use it to subset data based on a condition.

40

Explain the difference between `find()` and `filter()` when working with arrays.

MODEL ANSWER

`find()` returns the first element matching a condition (or undefined), while `filter()` returns all matches as an array. Use `find()` when you need one element: `links.find(link => link.id === 5)`; use `filter()` when you need all matches: `links.filter(link => link.id > 5)`.

41

What is event delegation, and why is it useful when creating dynamic lists of elements?

Think about efficiency and handling events for elements added later.

MODEL ANSWER

Event delegation uses a single listener on a parent to handle events from child elements. When a child fires an event, it bubbles up to the parent's listener, which checks `event.target` to identify which child triggered it. This is efficient for large dynamic lists and avoids adding listeners to each element.

42

How would you remove a card element from the DOM when its 'X' button is clicked, using event delegation?

MODEL ANSWER

Use `event.target` to identify the clicked element, then remove it:
`document.getElementById('listContainer').addEventListener('click', (e) => { if (e.target.classList.contains('delete-btn')) { e.target.parentElement.remove(); } })`. This works for dynamically added elements without rebinding.

43

Name two console methods you would use to debug your JavaScript code and explain what they do.

MODEL ANSWER

`console.log()` prints values; `console.table()` displays objects/arrays as tables; `console.warn()` and `console.error()` show warnings/errors with color coding. These help you track variable values and program flow, making debugging efficient and visual.

44

How do you set a breakpoint in the browser's developer tools, and why is this useful for debugging?

MODEL ANSWER

Open DevTools (F12), go to Sources, click the line number where you want to pause, and run your code. Execution stops at the breakpoint, letting you inspect variables, step through code, and understand execution flow. Useful for finding exactly where bugs occur.

45

Describe how you would plan the data flow in a small web app that lets users add, edit, and delete social links.

MODEL ANSWER

Plan data flow: User enters data in a form -> JavaScript validates and stores in an object/array -> Display renders the list -> Edit/delete handlers modify the data -> localStorage persists changes. Sketch this as a diagram showing functions and data movement.

46

During manual testing, you notice that clicking 'Add' sometimes adds two identical links. What steps would you take to debug and fix this?

Consider event handlers and form state.

MODEL ANSWER

The double-add bug likely occurs because the submit event fires but the form doesn't reset, or the button isn't prevented from default behavior. Check for `e.preventDefault()` in the submit handler and ensure `form.reset()` is called. Inspect browser DevTools to confirm each click.

47

How would you validate a URL input in your app and show an error message if it's invalid?

MODEL ANSWER

Validate URL format with a regex or URL constructor: `try { new URL(userInput); } catch { alert('Invalid URL'); }`. Show an error message dynamically: `if (!isValid) { errorDiv.textContent = 'Invalid URL'; } else { errorDiv.textContent = ''; }`.

48

In your mini-project, how would you implement a light/dark mode toggle using JavaScript?

MODEL ANSWER

Store the current mode (light/dark) in a variable or localStorage: `let isDarkMode = false`. Add a toggle button that flips this value and applies a CSS class: `document.body.classList.toggle('dark-mode')`. Update styles in CSS to change colors when the dark-mode class is present.

49

If you were building your Social Bio Link app, how could you use an AI assistant like ChatGPT or GitHub Copilot to help you write or improve your code? What would you check before using its suggestions?

Think about code generation, explanations, and reviewing output for accuracy.

MODEL ANSWER

Ask ChatGPT to refactor your code, explain errors, or generate boilerplate. Provide context (your code, the problem) and review the output for accuracy and best practices. AI accelerates coding but should not replace understanding; always validate suggestions against the requirements.

50

Integrative: How would you combine ES6 destructuring and advanced array methods to extract all URLs from an array of link objects?

Consider using `map()` with destructuring in the callback.

MODEL ANSWER

`const {title, url} = links.map(link => ({ title: link.title, url: link.url }));` extracts and returns title and url from each link. Or use reduce to aggregate data: `links.reduce((acc, link) => acc + link.clicks, 0)` sums clicks across all links.

51

Integrative: Describe how you would fetch data from an API and then display each item as a new DOM element on the page.

MODEL ANSWER

`fetch(apiUrl).then(res => res.json()).then(data => { data.forEach(item => { const div = document.createElement('div'); div.textContent = item.title; document.body.appendChild(div); }); })`. This fetches data and dynamically creates and displays an element for each item.

52

Integrative: How would you use classes, event listeners, and localStorage together to allow users to add and persist custom social links in your app?

MODEL ANSWER

Create a `UserProfile` class with methods to load/save to `localStorage`. Use `addEventListener` to bind edit/delete buttons. In the click handler, update the object, save with `localStorage.setItem`, and refresh the display. This combines classes, events, and persistence.